

HISTORICAL RESEARCH

* DEFINITION: - Historical research or historiography attempts to systematically recapture the complex nuances the people, meaning events and even ideas of the past that have influenced and shaped the present (Berg & Lurie, 2012 p 305)

Historical research relies on wide ranges on wide variety of sources, both primary and secondary including unpublished materials

* Primary Sources

- Eye witness accounts of events
- Can be oral or written testimony
- Found in public records and legal documents, minutes of meetings, corporate records, recordings, letters, diaries, journals, drawings
- located in university archives, libraries or privately run collections such as local historical society.

* Secondary Sources

- Can be oral or written
- Second hand accounts of events
- found in text books, encyclopedia, journal articles, news papers, biographies and other media such as films or tape recordings

* HISTORICAL RESEARCH STEPS

1. Identify an idea, topic or research question
2. Conduct a background literature review
3. Define the research idea and questions
4. Determine that historical methods will be the method used
5. Identify and locate primary and secondary data sources
6. Evaluate the authenticity and accuracy of source

of materials

7. Analyze the data and develop a narrative exposition of the findings.

* ADVANTAGES OF HISTORICAL RESEARCH

1. No other method provide us with an opportunity to find insights about past events.
2. Identifying key elements of past negative events can play a significant role in preventing certain similar circumstances. In other words, it can teach a history lesson.
3. It can result in an important discovery on a global scale.

4. A researcher is not involved with the event.

* DISADVANTAGES OF HISTORICAL RESEARCH

1. Sources may not be reliable and may represent misleading facts about the events.
2. Not all past events can be studied because of the lack of evidence and source materials. It gets more challenging to have research of the events are from a more distant past.
3. Finding a research for certain events does not mean that the same reason can lead to similar events. Because it may have happened in a totally different circumstance in the past.
4. The research may last longer, depending on the historical events and data availability.
5. May require travels between various destinations.
6. A researcher does not have control over the results.

NATURE AND TYPES OF RESEARCH

1.1 LEARNING OBJECTIVES

- 1. Upon completion of this chapter, readers should be able to:
 - 1.1 define research in specific terms.
 - 1.2 identify the purpose, functions and benefits of research in accounting and distinguish the major types of research.

1.1 INTRODUCTION:

Research has been defined by various writers to suit the circumstances in which it is used. Literally, the term research means 'to search for'. However, from the different definitions, it is seen as science due to its characteristic procedures involving logic of inquiry about the truth of an event or object. In view of this, research is a systematic investigation of a particular phenomenon, using a set of procedures, which is embodied in the scientific methods of presentation. Since research is basically science oriented, it may be applied or carried out in an informal setting for societal use.

Thus, the attributes of scientific approach of conducting research will include;

- ◆ Logical systematic presentation of procedures.
- ◆ Ability to control the variables involved in the studies.
- ◆ Information or data collected must be empirical that are based on observational facts or experience alone.
- ◆ A critical analogy of the hypothetical assumptions made about the variables that are being investigated.

Definition of Research:

Asika, (1991) defines research as any organized enquiry that aims at providing information for solving identified problems. In the same vein, Fagbohunge (1993) defines research as a quest for new knowledge pertinent to an identified area of interest or problem through the application of the scientific approach or process of investigation.

In less restricted sense, however, research refers to an organized investigation of a problem, a systematic approach to its solution or alleviation of the simple,

complex, immediate or long-term problem. Howard and Sharp (1983) view research as 'seeking through methodical processes to add to one's own body of knowledge and hopefully, to that of others, by the discovery of non-trivial facts and insights'.

A pertinent question that may be asked however, is 'what is the role of research in communication?' According to Reinard (1994), research is the systematic effort to secure answers to questions. He terms it as a very practical effort to get answers to questions, and it usually requires examining past inquiry into the issue at stake. On the other hand, Osuagwu (1999) defines business research as a study conducted or a systematic and objective method of finding solution to problems relating to the business and the environment.

Thus, business research has been seen as a systematic and objective collection, analysis, interpretation and reporting of data and information for business decision-making.

An empirical cycle is derived and shown as follows:

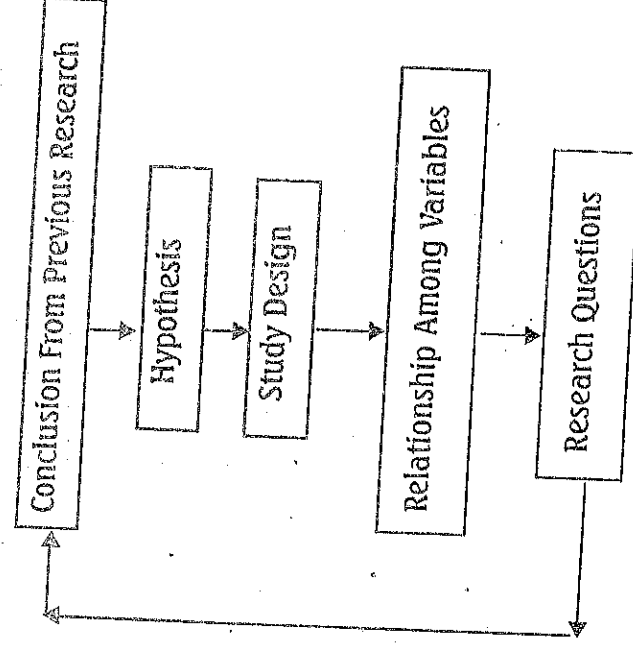


Figure 1: Research Cycle 1

Source: Sannie, M.B.A (2001)

From the various definitions of research given above, research is simply the process of arriving at dependable solutions to problems through a planned and systematic procedure which involves collection of data, analysis and its interpretation. It is the most important tool for advancing knowledge, for promoting progress and for enabling man to relate more effectively with his environment, so as to accomplish his purposes and to resolve his conflicts. Research is, therefore, directed towards the discovery of the relationship that exists among phenomena of the world in which we live.

1.2. THE PURPOSE OF RESEARCH:

(a)

The purpose of research can be deduced from most of the definitions proffered. This can be achieved by asking the question, 'why is the need for research?' A brief explanation to the question is summarized as follows;

- (i) To quench curiosity.
- (ii) Willingness to solve a problem.
- (iii) Desire to find out the relationships that exist between objects or concepts; that is, variables.
- (iv) Willingness to reduce the level of uncertainty when plans are being made.
- (v) Decision or intention to monitor performance after the plans had been put into operation.

Furthermore, the various classifications reveal the purpose of research. This is usually revealed in the goal of the type of study.

In general, research inter-alia,

- (i) Is directed towards the situation of a problem, that is, it is concerned with finding the truth about an event or object.
- (ii) Is involved in the quest for answers to problems. The process involves a high degree of originality.
- (iii) Is involved in understanding the events in the environment. Better understanding or knowledge of events gives room for better prediction and control.
- (iv) Enables individuals and organizations to be able to control events in the environments. A typical research cycle is shown as follows:

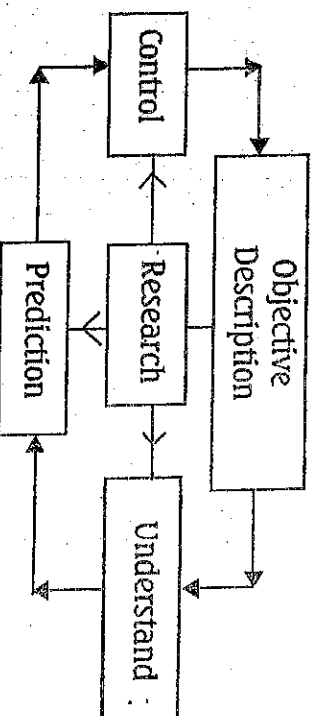


Figure 2: Research Cycle 2

Source: FagbohunGb (1993:3)

Further insight into the general purpose of research includes the fact that it:

- ◆ helps to make prediction of future events. An individual or organization that can predict future environmental changes has better chances of survival than one that lacks such ability.
- ◆ is concerned with objective description of events in the observable world.
- ◆ involves a careful collection, organization and articulation of what is already known about the problems and what is yet to be known.
- ◆ is a structured process, which follows a systematic order or rule of execution. Such rules include the procedure for identifying and defining variables, establishing the plan of study and relating the data thus collected to the originally stated problem and hypothesis.
- ◆ is characterized by carefully designed procedures that apply how the steps for data collection are followed; the subjects or objects studied; how the instruments are administered or the observations made and recorded, how the data are analyzed.
- ◆ involves rigorous application of logic and objectivity in the procedures employed in the data collected and conclusions reached.
- ◆ is based upon observable experience or empirical evidence; thus rejecting revelation and dogma as method of establishing knowledge.
- ◆ demands accurate observation and description. To achieve this, the researchers devise or select valid data gathering instruments and psychometric devices to refine observations, description and data analysis.

(b)

The Purpose of Research in Accounting

According to Inanga in Anao and Uche (2002:9-10), the place of research in the accounting practice has no linkage and as such the relationship is said to be disjointed. This, they identified, has resulted from lack of conceptualization of the primary objectives of the study of accounting, which in time past, has viewed the accounting practice in terms of information preparation only. However, the reaction of stakeholders such as investors, creditors and others to the need for historical facts and figures in terms of profit made on investment and revenue generation and analysis to be published, has geared the Accountants to be involved more in research. Bearing this inter-

relationship in mind, the researcher should select and expand his research questions with a view to controlling the empirical system within which accounting is practised. Research, according to Inanga, aids in improving the behaviour of the practice and makes it relevant to the society. More so, research will help to develop the accounting standards of system in such a way that it will move towards intended goals.

1.3 IMPORTANCE, USEFULNESS AND BENEFITS OF RESEARCH

The purpose of research provides an insight into the importance, usefulness and benefits derived from conducting research. Again, it depends on the situation in which the study is located. For instance, research can be applied to functional areas of business such as accounting, finance, marketing, personnel, administration, production and so on. Operations research has been used extensively in finance and production functions and other areas of business administration. Thus, research is used specifically:

- (a) To study cash flow, cash budgeting and investment analysis.
- (b) For financial planning and forecasting, that is determining operational changes and growth in cost-benefit analysis, capital budgeting, credits and collection policies as well as their effects on cash flow.
- (c) To identify markets, analyse buyers' behaviours and attitudes, gather sales data, data on market shares and examine market-potentials of a firm.
- (d) For the recruitment and selection of new employees.
- (e) For the preparation of job analysis and job evaluation.
- (f) In industrial relations and workers' compensations and benefits.
- (g) To develop solid and concrete evidence for socio-economic and political public policies.
- (h) To formulate and coordinate management information system, which in recent times, has been of huge success both in collecting and marketing data and information. These data are gathered, analysed and reported through marketing research process.

1.4 TYPES OF RESEARCH

Different writers have different opinions on the method of presentation of the various types of research. While a consensus has been reached, we intend to streamline the various positions to suit the readers' understanding.

Before we identify the types of research, we shall examine briefly the areas that research study can cover. This is so as not to confuse the types of research with the areas studied in research as we go on in our study. Research in itself covers a wide scope, such as;

(a) Action Research

This seeks to find solution to problems in a particular setting, where it can be applied. For instance, a study of the opinion of manufacturers and consumers to the ban on foreign products.

(b) Operations research

This involves quantifying a problem and developing mathematical models that can be used to explain such events for decision-making.

(c) Market research

This aims at discovering the needs of consumers.

(d) Materials research

This aims at improving the quality of raw materials used in a manufacturing process.

(e) Manufacturing research

This seeks to develop tools and equipment as well as methods of improving manufacture so as to reduce the cost of production and simultaneously increase productivity and profit margin.

(f) Product research

It seeks to develop the technology that is used in the production of old products and develops new techniques for the improvement of such products so as to meet the diverse needs of the consumers.

(g) Applied research

Applied research addresses practical problems and seeks to find potential solutions to them. It is concerned with testing and applying established theories, relationships and evaluating the usefulness of the theory in solving problems.

(h) Basic (pure) research

This is basically a fact-finding research. It tries to find answers to theoretical issues in a given situation under study for the purpose of knowledge acquisition. Questions such as, 'what is the effect of the study on the proposed goals and objectives?' or 'what are the factors affecting the outcome of the study?' are answered. This type of study leads to the development of theory, which may result in finding solutions to persistent problems. For instance, the effect of the ₦ 25 billion capital base in the Nigerian banking industry on financial

sectors in the country. Financial market analysts have come up with the fact that this policy has yielded positive result recently in that it has aided some financial sectors to rise from their poor status to a fair financial status in terms of revenue generation and payment of dividends to investors.

(i) Evaluation research

This makes use of facts collected, analysed and interpreted for decision-making. In order to achieve success in this research, the cost and the sustainability of the programs are considered. Questions such as, 'Is the programme meeting the set goals and objectives?' or 'Are there enough facilities to enhance the implementation of the programme?' are answered.

1.5 CLASSIFICATION OF RESEARCH BY METHODS - SPECIAL CASES IN BEHAVIOURAL RESEARCH

Research can be sorted into two main types; these are basic (pure) research and applied. The characteristics of these have been mentioned above. The other areas of research can be classified under these two types, and these may be further subdivided according to their procedure or method of investigation. This will form our guide for the discussion of this aspect of our study.

Descriptive research

Descriptive research allows the assessment of certain attributes, properties or characteristics in a situation prevailing at a particular time. Its main purpose is to describe the event in question using the resulting data to explain and predict the given situation. It gathers consistently the date of occurrence of events, which are used to give explanation of the relationship, test hypotheses, make predictions or get meaning and implication of the situation. Its main characteristics include the relationship between variables, testing of hypotheses and the development of generalization principles or theories that have universal validity. It can be classified as basic research with a qualitative characteristic where the elements in research are subjective in nature. Here, the researcher is free to express his opinion on the situation in view. Descriptive research explains a given phenomena by tracing what is already in existence and its effect on the variable(s) to be studied. Consideration of events, which had occurred in the past and are of relative importance to the present situation, further enhances this type of research. The researcher who embarks on descriptive study views the influences encountered from previous studies as they relate to present conditions that are under study. For instance, a study of the effect of the persistent increase in the pump price of petroleum products on the standard of living of middle and low-income earners in the last five years reveals the following explanations.

The expectation is that if variable A (increase in pump price of petroleum product) is systematically related to or affects variables B & C (standard of living of the middle and low income earners), then prediction of the future situation is possible. The result may suggest additional or competing hypotheses to be tested. The researcher is not just going to get the price to be increased for the purpose of investigation; he takes a look at the increase over the years and relates this to the standard of living of the group in question. He uses this finding to describe what was obtainable before and after each increase and discovers the effect on the living conditions of the affected citizens.

Historical research

This is a type of descriptive research in the sense that it has the same characteristics of being subjective in nature. It however differs from it in that it takes a look at past events in order to understand the present and then predict the future with such findings. Its major focus is on the record of dates of events and its influence at the time of occurrence. Its findings and conclusion are also subject to the researcher's opinion. For instance, the influence of World War II on the world's economic development of a nation.

Survey research

This gives a critical examination of events, objects, subjects or ideas with the intent of providing the exact information about the conditions of such situation. The method investigates problems, which are usually viewed with large or vast population sample. If the population is a large one, those who to participate in the study are randomly selected. Thus, the survey is called sample survey. In a case where the entire population is involved in the investigation, it is called census survey. The survey under investigation has direct influence on the variables involved.

Survey research can be qualitative or quantitative in character. Qualitative in that the findings of the study are based on the researcher's opinion. That is, it is subjective. It is also quantitative in the sense that the procedure may involve the collection of data and analysis, which is objective in approach. For instance, an opinion poll conducted among the Nigerian labour force on the policy of monetization and its effect on civil servants in Nigeria.

Case-study research

This focuses on a particular event, person or situation. The period of investigation is defined and its conclusion is confined to the study in view. That is, the result obtained here is used only for the situation and population that are investigated. For instance, the importance of locally manufactured goods on the Nigerian economic development during the era of democracy. The study above specifies the period as well as the population group to be studied. This is the major characteristic of this type of research.

Co-relational research

This type of research seeks to establish relationships between the variables or levels of association between the variables. It seeks to determine if the relationship that exists is significant and also the nature of the relationship, whether it is positive or negative. It has the nature of applied survey – thus, it can be categorized as being quantitative or qualitative. For instance, the relationship that exists between managerial staff and their sub-ordinates as it affects organizational productivity.

Causal comparative or ex-post-facto research

This determines cause and effect relationships by examining the conditions that are traceable to the probable causal factor of the event under investigation. The effect and cause have already occurred but these conditions are studied in retrospect. It starts by identifying the effect after which, the cause is then determined. This is because the effect is already noticeable in the population under investigation; however the researcher will carry out a further study so as to prove this, after which he carries out another or further studies to determine the cause of the situation under investigation.

It has the descriptive characteristics and as such can be either qualitative or quantitative. Quantitative in the sense that the findings and conclusions are specific; thus objective in its finding.

Experimental research

Hassan (1995) defines experimental research as a scientific investigation in which, the researcher manipulates, controls and describes one or more independent variable(s) and observes the effect on the dependent variables. The major feature of this research type is that it has a control or manipulated group. This control or manipulated group could have features such as that of descriptive research.

Experimental research however, investigates the possible cause-and-effect relationship between two or more groups. This is achieved by exposing the experimental group (that is, the group to be studied) to one or more treatment, and comparing the result to that of the control group (the group that does not receive any treatment or the placebo group). The treatment applied to the experimental group is applied so as to vary certain aspects of the situation under study. The control group on the other hand, aims at determining the exact position or behaviour of the subject so as to have the knowledge of the antecedent condition of the situation, which invariably helps the researcher to compare the two situations. The effect of manipulation is achieved by using one or more descriptive methods. This is done by determining the states of certain relevant characteristics, which are dependent or independent variables.

Experimental research is divided into two: true and quasi-experimental research.

In true experimental research, the researcher manipulates at least one of the variables of study – the independent variable; controls other relevant variables that can produce effects which are similar to the independent (dependent variable) and then observes the effect of the change on another variable.

Another unique feature of true experimental study is that it concentrates on the control of extraneous variables; these are factors that are not necessarily part of the study but which may influence the result of the study, and the randomization of subjects from the population of study. This aims at determining the internal and external validity of the research outcome.

The quasi-experimental research, on the other hand, takes into consideration the influence of those variables (usually extraneous) that can be partially controlled or totally uncontrollable. In this type of research, the study involves real-life setting, in which some subjects cannot be assigned treatment; neither is it possible to control them because of the influence of the extraneous variables.

The quasi-experimental design is thus used to separate the influence of the treatment condition. It meets all the experimental modes of analysis and data interpretation except for the control and randomization of subjects in the population of study.

A typical instance of quasi-experiment is fields experiment on the effect of the implementation of the monetization policy on the recipient and non-recipient civil servants in Lagos State.

1.6

RELATIONSHIP BETWEEN TYPES OF RESEARCH

Having examined the various types of research, it can be observed that most of the research types have related characteristics or properties and as such no one of them exists in isolation. They are therefore not completely independent. Some studies are inter-related and therefore have similar characteristics. For instance, inasmuch as descriptive research is subjective in terms of result analysis, it has the nature of establishing the relationship that exists between two or more variables.

Descriptive research is similar to the experimental research study in terms of establishing the relationships that occur between variables. There major differences lie in the fact that experimental study makes use of manipulated and controlled variables as well as total randomization of the samples, which are involved in the study except for extraneous cases (quasi-experimental) inter-woven - a degree of description is done in experimental while descriptive research does not make use of manipulated and controlled variables. More

so, like historic research, which gathers past information to determine the present and predict the future, descriptive research also gathers past information, which it uses to determine the present effect thereby predicting the future phenomena.

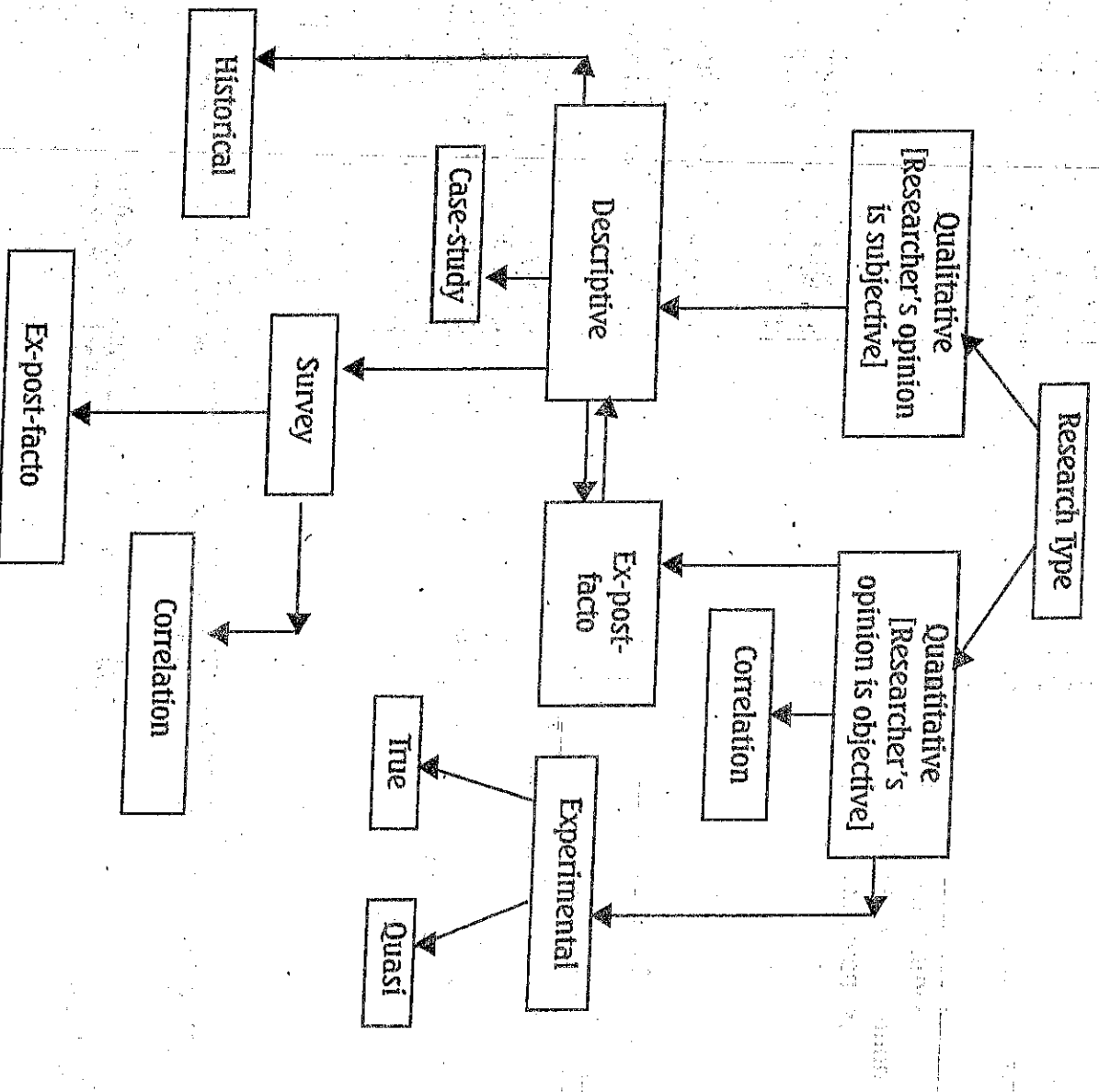


Figure 3: Classification of Research Types by Method of Study
Source: Yinke-Segilola (2004)

STATISTICAL ANALYSIS AND PRESENTATION

8.0

LEARNING OBJECTIVES

- On the completion of this chapter, the reader will be able to
 - define what is associated with this chapter's main summary tables, graphs, charts and statistics
 - present data as grouped or ungrouped, and all the forms mentioned above
 - interpret information from the graphical presentations
 - identify the methods of statistical data analysis (descriptive, inferential and computer) and
 - identify and solve examples involving measure of central tendencies

8.1

INTRODUCTION

This chapter deals with statistical data summary and analysis. It examines the use of tables, charts and graphs to present data in an easier fashion to interpret. Discussion on the concepts of descriptive analysis is given and detailed explanation on measures of central tendency and relationship is highlighted with illustrative examples.

8.2

DATA SUMMARY

After collecting data from various sources, the data are then analyzed using appropriate statistical methods.

Data summary however, is a presentation of data, and how they are collected and the forms the data take as well as the methods adopted for the analysis. This aspect is a prelude to full discussion on the data collection procedure, interpretation and findings of the results. The data summary is, therefore, the presentation of data in tables, charts and graphs.

It, however, involves presentation and analysis of data in a concise form for clear understanding of the reader and the user of the data.

◆ Tables

Tables are one form of coding data which are presented in clearer form for easy understanding. There is simple tabulation of data and it may be necessary where there are small numbers of cases. In this case, the larger summary sheets containing data from all cases are sometimes useful for quick overviews of the data.

The simplest form of representing research data, is the frequency distribution tabulation. All that is meant by this is the presentation of information or data in one column of different qualities of an attributes or different values of a variable, together with the entries in another column showing the frequency of the occurrence of each of the attributes or classes of attributes. For instance, a summary of frequency distribution can be presented in tabular form. This makes the statistical measures such as central tendency or dispersion easier to be analyzed.

The following is a table illustrating the use of ratio for comparing frequencies.

Death Ratio over Age and Sex of Population per Thousands

Age	Male	Female	Ratio of Male to Female
0 - 4	17.2	13.6	1.26
5 - 9	2.4	1.7	1.41
10 - 14	1.5	1.2	1.25
15 - 19	2.4	1.9	1.26
45 - 54	12.3	10.0	1.23
55 - 64	25.8	19.9	1.30
65 - 74	59.1	45.9	1.29
75 and over	135.6	114.5	1.18

Table 8.1: Data Presentation of Population by Age and Sex in Two Dimensional Forms

Source: Methods In Social Research by William, J.G. and Paul, K.H. (1981:346)

The above, is a two-dimensional table, the pieces of data are presented in two directions. The two variables of male and female are being

compared under two separate columns in relation to their age, while other variables in the first column are shown horizontally with the ratio of the genders appearing in the last column.

However, a one-dimensional table is just a case of one independent versus one or more dependent variables. This is represented below:

Dichotomy	Responses
Yes	45
No	36
Undecided	19
Total	100%

Table 8.2: Data Presentation of Responses in One Dimensional Form

Source: Thurnstone postulation

Charts

The charts and other diagrams can be used to present figures in the form of data. This is an aspect of non-verbal communication. Aside from the use of words, they use a form of symbols.

Charts are basic aids in translating complex statistical tabulations into a form which can be grasped easily. These consist of bar charts, pie charts, picture charts or pictogram- non-qualitative in nature. For instance, an organizational chart depicting the relationship between various levels of executive and staff interactions or status and the functions performed by the members of staff in the company or firm. The bar, pie and picture charts are still regarded as crude methods of presenting data in a readable form to help the reader understand the more complex tabulations, which are given.

Examples of these types of chart are illustrated below.

Year	Subscribed Asset	Accrued
1991	150	100
1992	250	200
1993	350	280

Table 8.3: Table of Price of Commodity Showing the Subscribed Asset and Accrued Liability for a Period of Three Years.

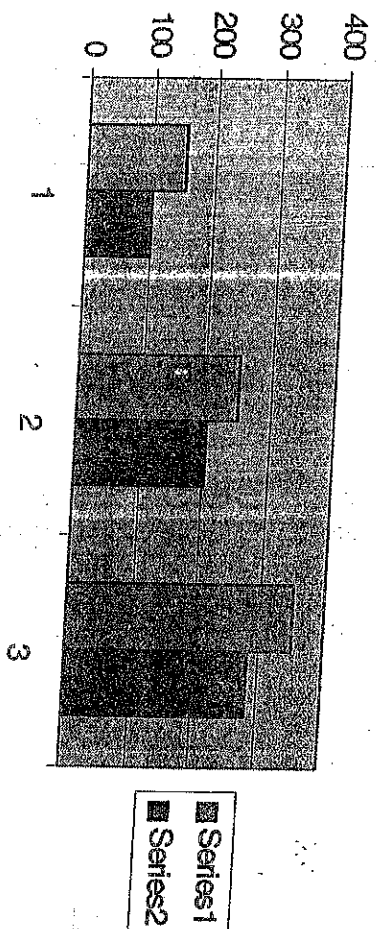
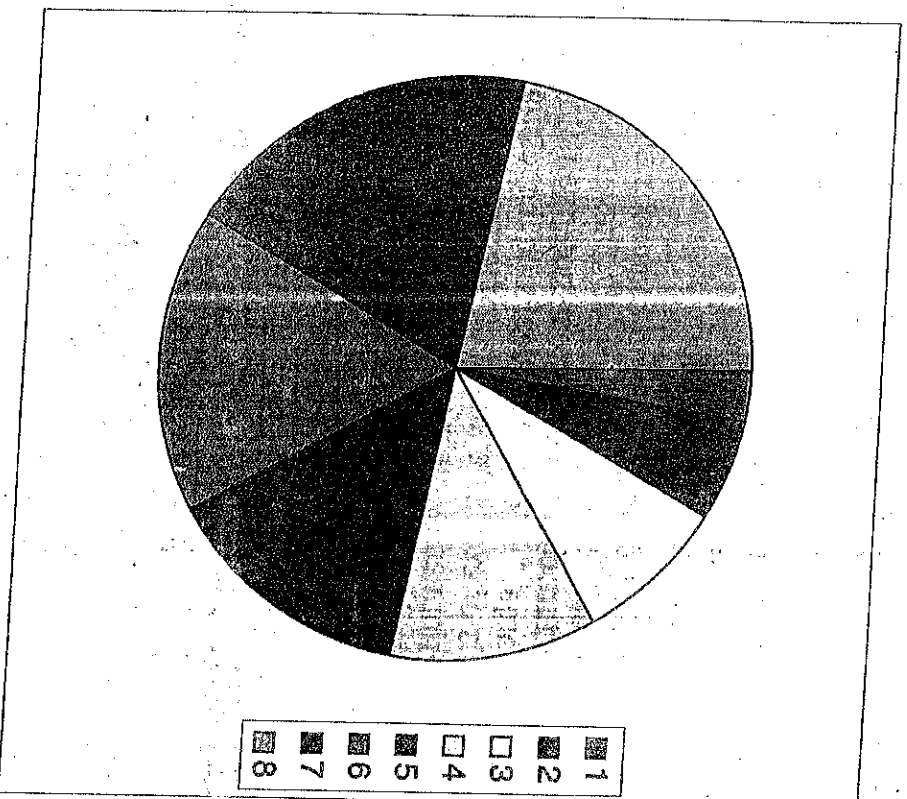


Figure 8.1: Dual Bar Chart

Figure 8.2:

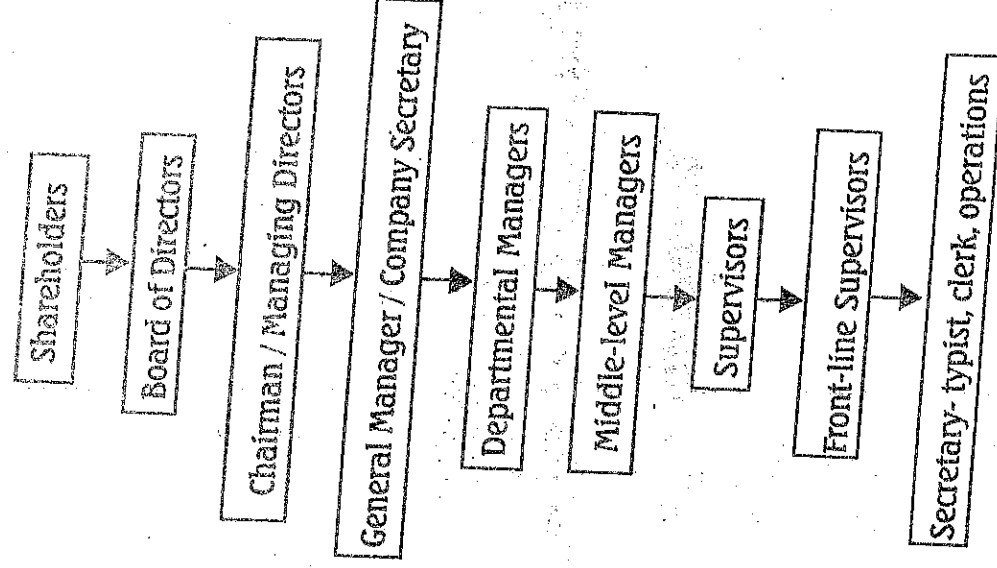
Pie Chart



Key:

S/N	Key
1	Advertising- 30%
2	Public relation 15%
3	Publicity 10%
4	Packaging 25%
5	Sales promotion 5%
6	Personal selling / direct marketing 5%
7	Event marketing 5%
8	Other types 5%

Figure. 8.3: Organizational Chart- A Pyramid of Vertical Communication Structure



◆ Graphs

Graphs are also visual aids that are used in presenting data and translating statistical tabulations for easy understanding of the readers. The graph is one of the most common devices for proving a trend. Graphs, which show trend, can be most accurately described by mathematical equations.

Graphs are pictorial presentation of statistical data and are normally plotted to indicate associations between variables. Some examples of graphs are:

- ◆ Straight-line graphs
- ◆ Trend graphs
- ◆ Lorenz Curve, and so on.

The following illustrations show the relationship between the prices of a product over the last few years against the demand during the last few years.

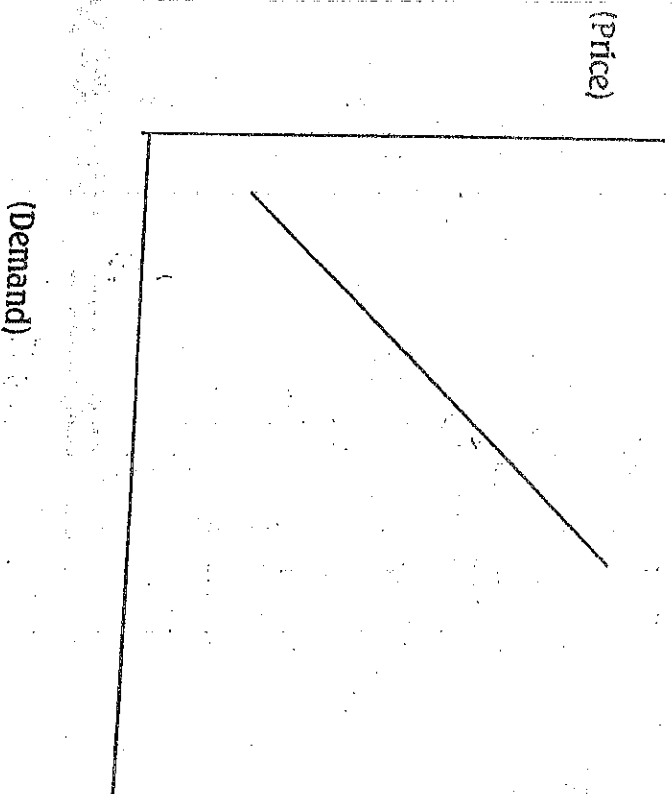


Figure 8.4: Perfect Positive Demand Graph

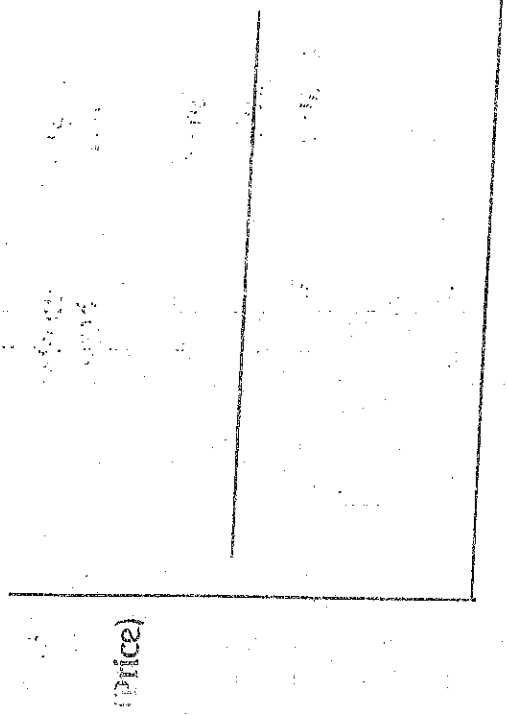


Figure 8.5: No relationship in demand and price graph

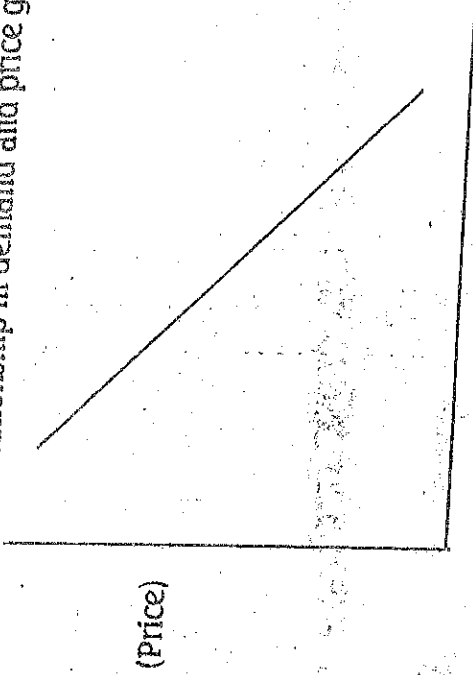


Figure 8.6: Perfect Negative Demand Graph

8.3

SUMMARY STATISTIC

Statistic(s) is divided into two types. These are:

- I. Descriptive Statistics, and
- II. Inferential Statistics.

The summary statistic is that aspect of statistic that estimates or draws a sample from the total population or universe and makes generalization based on the conclusion drawn there from. That is, it is the summarization of the data by making value judgment. In this case, certain inferences are made. Inferential Statistics help us to infer from these statistical measures of central tendency and dispersion, certain meanings and important relationship, which lie hidden within the data. The main functions of inferential statistics are to:

- Estimate a population parameter from a representative sample drawn from the population.
- Predict the population characteristics from the characteristics of a randomly selected sample or samples of the population.
- Test relevant hypotheses for the purpose of drawing valid conclusions for research studies.

There are two divisions within the realms of inferential statistics that give credence to statistical testing. These comprise parametric and non-parametric statistics.

The former includes the mean and its components, Standard Deviation, Correlation Analysis, Regression Analysis, Analysis of Variance (ANOVA), and other means of estimating the data gathered from the sampled population, which will be discussed fully in later chapters.

The latter is distribution-free statistics in which the tests conducted are not based on population parameters but can be valid under general statistical independent and homogenous variables. Spearman's rank order correlation and Spearman's rho (σ). Other types are Mann-Whitney U-Test, Kruskal-Wallis Test, The Wilcoxon Matched Pairs, Signed Rank Test, The Wilcoxon Rank Sum Test, Kolmogorov-Smirnov Test, Kendall Tau (Coefficient of concordance), Corner Test, Media Test and Sign Test.

8.4

STATISTICAL TOOLS IN RESEARCH

What is Statistics?

Typical expressions such as, 'the representative manufacturing company...', 'the average family income,' or average production capacity' have been said or heard over times in our daily routine. All these and many more are referred

to as statistical concepts. As group characteristics, they may be expressed in measurement of size, sex, age, weight and other traits that can be described quantitatively. When we say that the average income per family in Nigeria is Fifty Thousand Naira (N50, 000.00). We are generalising about all families in Nigeria and not about a particular family. Thus, this signifies an abstraction that may be used in place of great mass of individual measures and as such is referred to as statistical measurement.

The use of statistics in a research project is much more than just the manipulation of data. It serves the fundamental purpose of description and analysis. Its proper application involves getting the following attributes as identified by Best and Kahn (1989:221) in place:

- The ideas that are needed to gather the data for answering research questions and to test the hypotheses.
- The methods of selection, gathering, organisation and analysing these data.
- The assumption underlying the statistical methodology to be used.
- The generalisation of the conclusion drawn from data analysis.

In the light of the above, we can conveniently define statistics as a quantitative process of gathering, organising, analysing and interpreting numeric and non-numeric data for the purpose of making a general application, which may be used to explain the phenomenon at hand and to predict the future occurrence.

8.5

METHODS OF STATISTICAL DATA ANALYSIS

In times past, two main methods of statistical data analysis have been measured and used by researchers; these are descriptive and inferential data analysis.

Recently, the use of computer data analysis is gaining ground and most researchers now use statistical computer packages for data analysis. These three concepts shall form our basis for discussion in this part of the study.

8.5.1 Descriptive Data Analysis

This shows mainly the tabular and graphical presentation of data as explained previously. It deals with the classification of data. Its major characteristics are the fact that it limits generalization to the particular group observed and conclusions are not extended to other groups besides the observed group. Thus, it provides valuable information about such group.

Examples of statistical methods here include; tabular presentations such as frequency distribution, which results from data classification

and graphical presentations such as histogram, bar charts, cumulative frequency curve (Ogive) and pie or pictograms. Other basic types of statistical measures used are:

Measure of central tendencies (average) - mean and its components mode and median.

measure of dispersion/spread (variation) – range, variance, absolute mean deviation, coefficient of variance and skewness (Normal Distribution).

Measure of relationship – coefficient of correlation of various types for instance; rank, Spearman and Pearson's.

quantiles, deciles and percentiles, standard scores and others.

8.5.2 Frequency Distribution

Frequency is the number of occurrence of an event. Thus, its distribution is presented in a table of either grouped or ungrouped figures showing the variables and the number of items that they occur.

ILLUSTRATION 8.1

The following raw data are credits of 50 candidates who sat for the 2002 May ICAN examination.

3 2 2 1 1 2 3 3 2 1
1 2 2 3 3 1 2 2 3 1
1 1 1 3 2 3 1 1 3 3
2 3 3 1 3 3 2 3 1 2
2 1 2 2 3 2 3 3 2 1

The data above have not been treated in any form, thus they are referred to as raw data scores.

When the data are arranged in order of size of either ascending or descending order, then we have an array of data. The data above can be represented as an array of ascending order data as follows.

I I I I I
 I I I I I
 I I I I I
 N N N N N
 N N N N N
 N N N N N
 W W W W W
 W W W W W
 W W W W W
 W W W W W

Table 8.4: Table Showing the Credits of 50 Candidates in the May-June 2002 ICAN Examination

i. Ungrouped Frequency Distribution

This is a table that shows, in a single file, all the figures that appear in a raw data. (Irrespective of the number of occurrence) The frequency is placed along side, and it shows the number of occurrence of each figure. For instance, from the table above, we can form an ungrouped frequency distribution table.

CREDIT	FREQUENCY (f)
1	15
2	17
3	18
	$\Sigma(f) = 50$

Table 8.5: Ungrouped Frequency of the Credit of 50 Candidates in the May-Diet 2002 ICAN Examination

Where $\Sigma(f)$ is the sum of frequency of (f).

ii. Grouped Frequency Distribution

ILLUSTRATION 8.2

This shows a table of figures that have already been put into a collective unit that is grouped. This group or class of collection unit is called interval. For instance, if the marks obtained from a questionnaire administered to 50 students on the opinion about ICAN examination are as follows in (%)

72	84	55	96	47	85	37	73	54	66
39	66	55	55	53	50	70	91	72	90
85	85	68	75	81	93	47	60	58	75
54	40	42	70	63	62	61	80	79	55
74	75	90	95	72	80	67	66	43	38

Table 8.6: Marks obtained from the Opinion of 50 Students about ICAN Examination (ungrouped data)

In order to distribute this raw data into groups, a tally is used, that is, a slash, whenever a corresponding mark is reached as obtained below. The following steps are taken, when grouping a raw data.

- ◆ Determine the range i.e. the difference between the highest and lowest scores
- ◆ Determine the required interval and divide the range by the interval. This gives the 'class unit' (c) represented by an interval.

NB: If a decimal occurs in your division, an additional class is provided. Thus for the raw data above we have:

- a. Range = $96 - 37 = 59$
 b. $59 \div 6 = 9.8 \sim 10$
 Thus the class unit (c) is 10

CLASS OF MARKS	TALLY	FREQUENCY (f)
37 - 46		6
47 - 56		9
57 - 66		9
67 - 76		12
77 - 86		8
87 - 96		6
		$\Sigma (f) = 50$

Table 8.7 Table of Frequency Distribution of Marks obtained from the Opinion of 50 Students about ICAN Examination

NB: In order to check if the class interval, tally and frequency are evenly distributed, the sum of frequencies (f) that is $\Sigma (f)$ must be equal to the total number of observations made during a study. In this case the number of marks obtained from the sampled population is equal to 50

iii. Statistical Terms Associated with Group Frequency

Class Interval: This is the range of each class; for instance 37 - 46 is the first class interval, 47 - 56 is the second-class interval, 57 - 66 is the third, and so on.

Class Limit / Boundary: These are numbers that mark the end of the class interval; for instance: the interval 37 - 46, has 37 as its lower class limit and 46 as its Upper class limit. What this implies is that in the lower class limit, the smallest number that can be listed is 36.5 and the highest number that can be listed is 46.5. Thus this phenomenon refers to the class boundaries for the interval.

Class Mark / Mid Point: This is the point halfway between the class limit and boundary of each Class interval. For instance, the second class of 47 - 56, the class Mark or mid-point is $47 + 56$ divided by 2, which is 51.5.

Size of Class: This is the difference between the lower and upper boundaries. For instance, in our example, the size of the class is 10, that is $46.5 - 36.5$. NB: The decimal is negligible.

8.5.4 Graphical Presentation of Data

These are diagrams representing the relationship between variables. The type of graph that is used for analysis is dependent on the nature of the data and the purpose for which it is intended. There are different types of statistical graphs and charts, but for this study, we shall consider the following graphical presentations;

- ◆ Bar charts.
- ◆ Histograms.
- ◆ Cumulative curves (Ogive)
- ◆ Pie chart.

Advantages:

- (i) Information that are obtained from a study can be coded and decoded at a glance.
- (ii) Reduces the boredom involved in reading write-ups.
- (iii) Gives better and closer view to the understanding of the data.
- (iv) Shows the relationship that exists between variables.

Disadvantages

- (i) It takes time to construct.
- (ii) It may appear cumbersome and difficult to interpret.

Bar Chart

A bar chart is a graphical presentation of the class of data which is represented by bars. The bars are not joined together and the length of each bar is proportional to the class frequency.

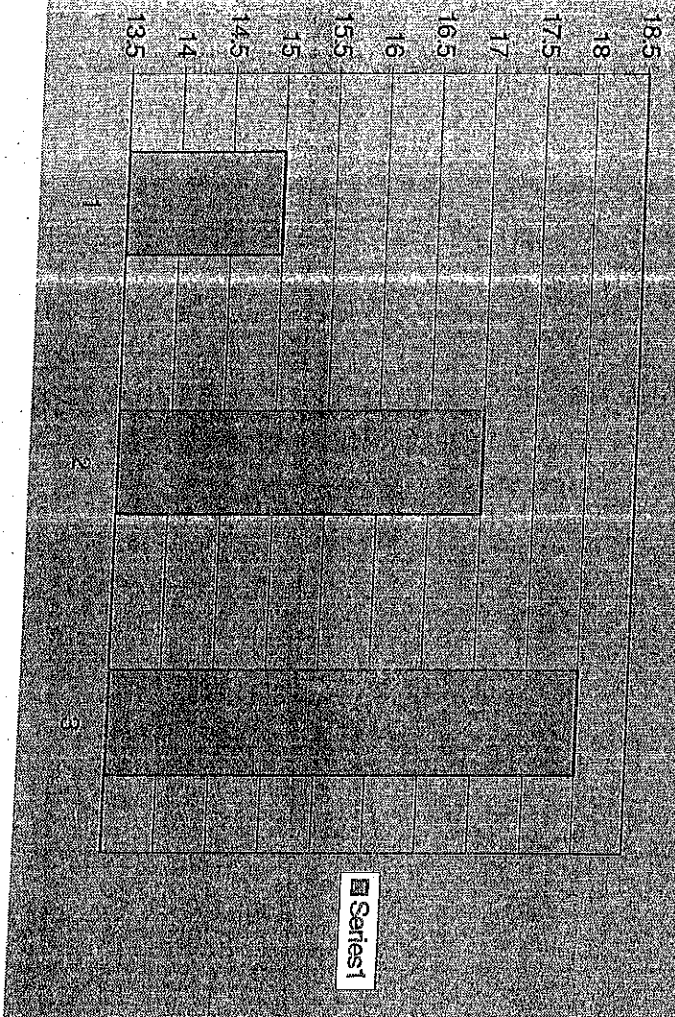
NB: The space between the bars and the width of the bars should be equal and of common size.

ILLUSTRATION 8.3

Given our example from Table 8.5-frequency table, a simple bar chart can be constructed as follows:

Credit (X)	Frequency (Y)
1	15
2	17
3	18

Figure 8.7 A Simple Bar Chart



Component or dual Bar Chart

This consists of two or more pieces of information obtained about a particular group within a period of time. It is used to compare the difference between the needed information, which is peculiar to those groups involved within a specified period of time. For instance, in an Institute, the employees are categorized into three groups as follows:

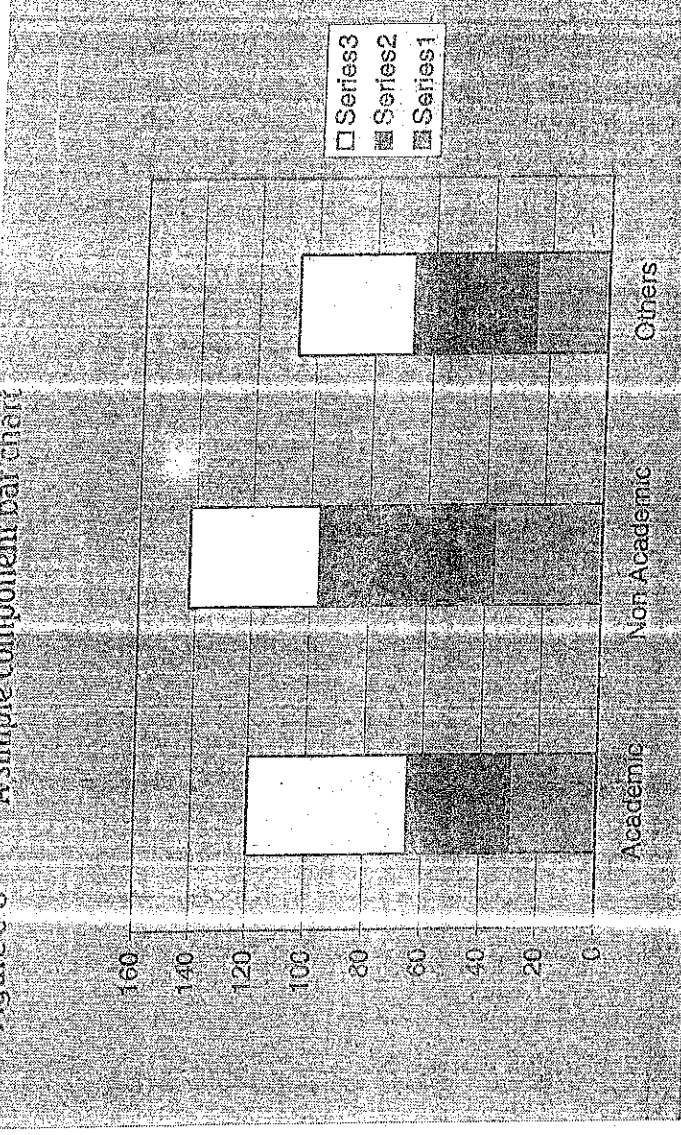
ILLUSTRATION 8.4

A Simple Component Bar Chart

Categories	A	B	C
Academic	30	35	55
Non Academic	37	60	45
Others	25	42	39

Table 8.8: The Categories of Three Groups of Employees in an Institution

Figure 8.8 A simple component bar chart



◆ Histogram

Unlike the bar charts, the vertical bars here are joined together. The height of bars represents the frequency. If an interval construction is to be done, then the class boundaries are used. As with the bar charts, the width of the bars is the same.

ILLUSTRATION 8.5

Using the class interval for test mark obtained from table 8.7, a histogram can be constructed as follows

CLASS OF MARKS	TALLY	FREQUENCY (f)
37 - 47		6
47 - 57		9
57 - 67		9
67 - 77		12
77 - 87		8
87 - 93		0
		$\Sigma (f) = 50$

Table 8.7 Table of Frequency Distribution of Marks obtained from the Opinion of 50 Students about ICAN Examination

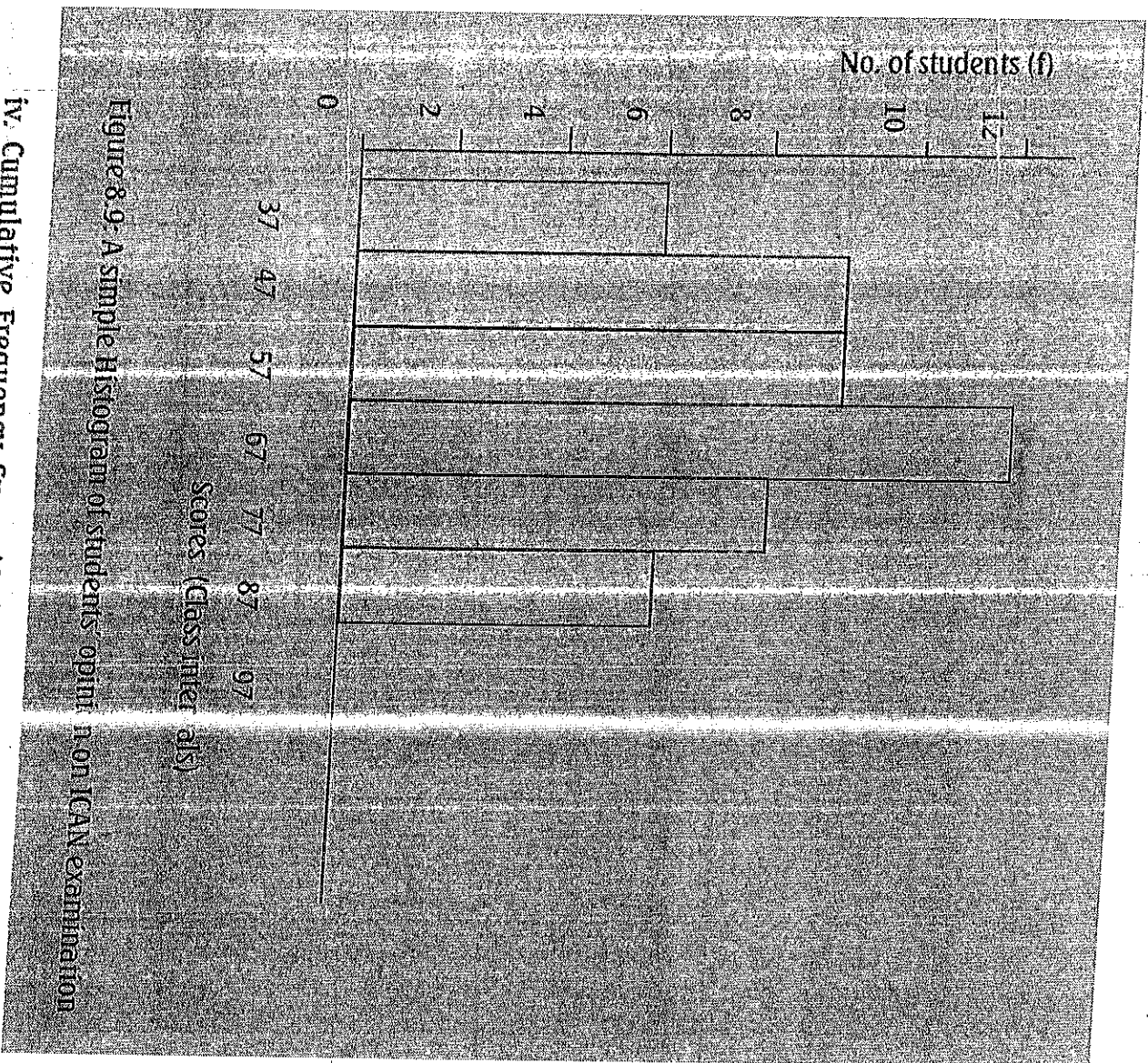


Figure 8.9: A Simple Histogram of students' opinion on ICAN examination

iv. Cumulative Frequency Curve (Ogive):

This describes the number of items that have values less than a particular level. It is obtained by adding continuously the frequency. It enables the observer to read the number of all the units directly. A graph of cumulative frequency distribution for a set of data is called Ogive or cumulative frequency curve.

ILLUSTRATION 8.6

The following observation is made among 30 factory workers in the month of October 2000. The table of observation is given below:

Time Interval (Month)	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60
Number of workers	3	5	10	8	4

Solution:

Time Interval (Months)	Number of workers (f)	Cumulative Frequency (c.f.)
10 - 20	3	3
20 - 30	5	8
30 - 40	10	18
40 - 50	8	26
50 - 60	4	30

Table 8.9 Table of Observation made among Factor Workers

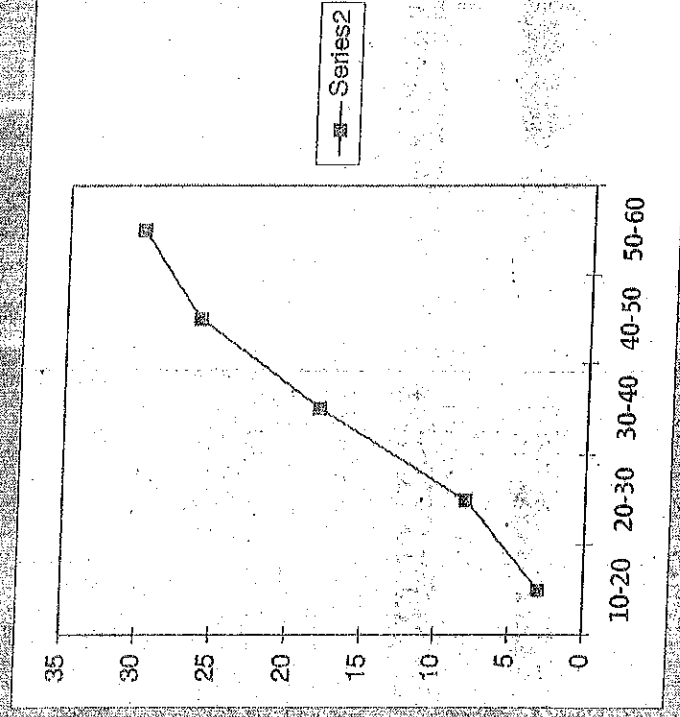


Figure 8.10: Cumulative Frequency Curve of Observation made among Factory Workers in October 2000.

8.6

MEASURES OF CENTRAL TENDENCY OR LOCATION

Central tendencies can be measured when the variables or data given concentrate at a mid-point. It can also be referred to as the average value. Examples include mean, mode and median which are commonly used in research analysis.

(a) Mean

A mean is simply the average of the observations or variables obtained. It is denoted by $\bar{X}$ and expressed mathematically as -

$$\bar{X} = \frac{\sum X_i}{N} \text{ ————— equation 8.1}$$

Where $\sum$ (sigma) is the summation

X_i is the i^{th} observation made.

N is the number of observations

ILLUSTRATION 8.7

The number of sales of product recorded in Naitira (N) by a departmental store during the months of January to December are recorded as follows. mean of the sales recorded for the year.

Solution:

Using equation 8.1.

$$\sum X = 10311$$

$$N = 12$$

$$\text{Thus } \bar{X} = \frac{10311}{12}$$

$$\bar{X} = N 859.25$$

Conclusion:

From the calculated average above, it implies that the average sales recorded by the departmental store for the year is N 895

Calculation of Mean from Frequency Distribution

For Grouped Data:

Data may be grouped when the value of X occurs more than once in a set of data. Therefore, the mean of such distribution is expressed as;

$$\bar{X} = \frac{\sum fx}{\sum f}$$

———— equation 8.2

Since the original data have been classed, we assume that the data in each group concentrate at a point called mid-point of each class of data. Thus, the mid-point represents the X -value along with their frequencies (f).

ILLUSTRATION 8.8

The weights of 30 students recorded during a physical test are as follows:

Class of Weight (pounds)	51-55	56-60	61-65	66-70	71-75
No. of students	3	6	10	3	8

SUGGESTED SOLUTION 8.8

Class of weight	Frequency (f)	Mid-point (X)	fx
51-55	3	53	159
56-60	6	58	348
61-65	10	63	630
66-70	3	68	204
71-75	8	73	584
	$\sum f = 30$		$\sum fx = 1925$

Table 8.10: Table of the Weights of 30 Students Recorded during a Physical Test

Since $\sum f = 30$

$$\sum fx = 1925$$

Using equation 8.2

$$\bar{X} = \frac{1925}{30}$$

$$\bar{X} = 64.17 \text{ (Usually taken to 2 decimal places)}$$

Conclusion

The average weight of the students recorded during the test is 64.17 pounds

ILLUSTRATION 8.9

For Ungrouped Data:

Given the number of observations recorded by a researcher on the level of performance of workers at the discharge of their duties for a number of x-
x_____10 9 8 7 6 5 4 3 2 1

f_____2 3 3 4 7 5 4 2 2 2

Solution:

x	f	fx
10	2	20
9	3	27
8	3	24
7	4	28
6	7	42
5	5	25
4	4	16
3	2	6
2	2	4
1	2	2
	$\Sigma f = 34$	$\Sigma fx = 194$

Table 8.11: Table showing the level of performance of Worker
Using equation 8.2

$$\bar{X} = \frac{194}{34}$$

$$\bar{X} = 5.71 \sim 6$$

Conclusion:

For a number of x- groups of workers, the average observation made on level of performance at the discharge of their duties is 6 observations.

Calculation of Mean Using Assumed Mean Method.

Here a number is taken from among the mid-point values (x). This number is usually taken from the highest frequency in the group or class of distribution. Thus to calculate this,

$$\bar{X} = A + \frac{\sum fd}{\sum f}$$

———— equation 8.3

Where $d = x - A$

ILLUSTRATION 8-10

Using the weight of 30 students recorded in the class interval above, with an assumed mean of $A = 63$, calculate the mean weight.

SUGGESTED SOLUTION 8-10

Class of weight	Frequency (f)	Mid-point (x)	$d = x - A$	fd
51 - 55	3	53	-10	-30
56 - 60	6	58	-5	-30
61 - 65	10	63	0	0
66 - 70	3	68	5	15
71 - 75	8	73	10	80
	$\sum f = 30$			$\sum fd = 35$

Using equation 8.3:

$$\bar{X} = 63 + \frac{35}{30}$$

$$\bar{X} = 63 + 1.67$$

$$\bar{X} = 64.67 \text{ pounds}$$

Conclusion:

The average weight of the students recorded during the physical test, using assumed mean is 64.67 pounds

When this result is compared with the one above, it will be noted that the difference is minimal.

(b) The Median

The median is the value of the middle item of a set of data, when the data have been arranged in order of magnitude; that is, in ascending or descending order

Thus, for a set of data with odd number of items, the median is simply the middle item after arranging the set of data in order of magnitude. For instance, for a set of ungrouped data;

4 6 7 9 10 13 15 26 35 42 45

The median is 13.

Thus, for a large set of data with odd number such as 105, the median will be

$$= \frac{105 + 1}{2} \text{ Item}$$

$$= \frac{106}{2}$$

$$= 53^{\text{rd}} \text{ item}$$

Thus the 53rd item after arranging the set of data in order of magnitude will be the median.

For even number item, the two middle numbers' average is taken thus if a set of data is given as;

2 3 5 7 9 11 13 16 19 20

Since the middle numbers are 9 and 11, then their average is,

$$\frac{9 + 11}{2} = 10$$

Therefore 10 is the middle observation of the ordered set of data.

Calculating the Median For Group Data

There are two major methods that are used in achieving this. These are;

- ◆ The interpolation, and
 - ◆ The graphical method.
- Interpolation Method:

The class that has the middle item is identified and the formula below is applied.

$$Me = l_1 + \frac{\frac{N}{2} - \Sigma fm}{f_{(med)}} \left\{ c \right. \text{--- equation 8.4}$$

where Me is the median,

L_1 is the lower class boundary of the Median class

N is the number of observation made

Σfm is the sum of frequencies lower than the Me class or simply the cumulative Frequency

c is the size of the Me class.

For a group distribution, the following procedure is adopted before using the formula above:

- ◆ Cumulative frequency (c.f.) is formed.
- ◆ Find the middle value by dividing N by 2
- ◆ Using the result in step 2, identify the median class. That is the class whose c.f. first exceeds the result in step 2.
- ◆ Identify the values that correspond to the symbols in the formula.
- ◆ Carry out or compute the calculation using the formula in equation 8.

ILLUSTRATION 8.11

Use the example 10 on the weights of 30 students recorded above to compute the median.

Solution:

Class	Frequency (f)	Cumulative frequency (c.f.)
51 - 55	3	3
56 - 60	6	9
61 - 65	10	19
66 - 70	3	22
71 - 75	8	30

The middle value is $\frac{30}{2} = 15$ th item

The median class is 61 - 65 that is the class whose c.f. first exceed 15 is 61 - 65.

Since $i_1 = 61$

$f_{(med)} = 10$

$C = 4$

$\Sigma fm = 9$

$N = 30$

Using equation 8.4:

$$\begin{aligned}
 Me &= 61 + \frac{30 - 9}{2} \times \frac{4}{10} \\
 &= 61 + 2.4 \\
 Me &= 63.4 \text{ pounds}
 \end{aligned}$$

ILLUSTRATION 8.12

A manufacturing company has the following results for the volume of bottles produced by process. Compute the median.

Volume of bottles	Frequency (f)	Cumulative frequency (cf)
485 - 490	1	1
490 - 495	3	4
495 - 500	12	16
500 - 505	22	38
505 - 510	8	46
510 - 515	2	48

Table 8.12: Table showing the Volume of Bottles per Day

Suggested Solution:

The median class is the 24th item (the class whose c.f. first exceeds 24th position) is 500 - 505

$$L_1 = 500, N = 48, \Sigma fm = 16, f_{\text{med}} = 22, c = 5$$

Using equation 8.4,

$$Me = 500 + \frac{48 - 16}{22} \times 5$$

$$Me = 500 + \frac{24 - 16}{22} \times 5$$

$$Me = 500 + [0.36] \times 5$$

$$Me = 500 + 1.818$$

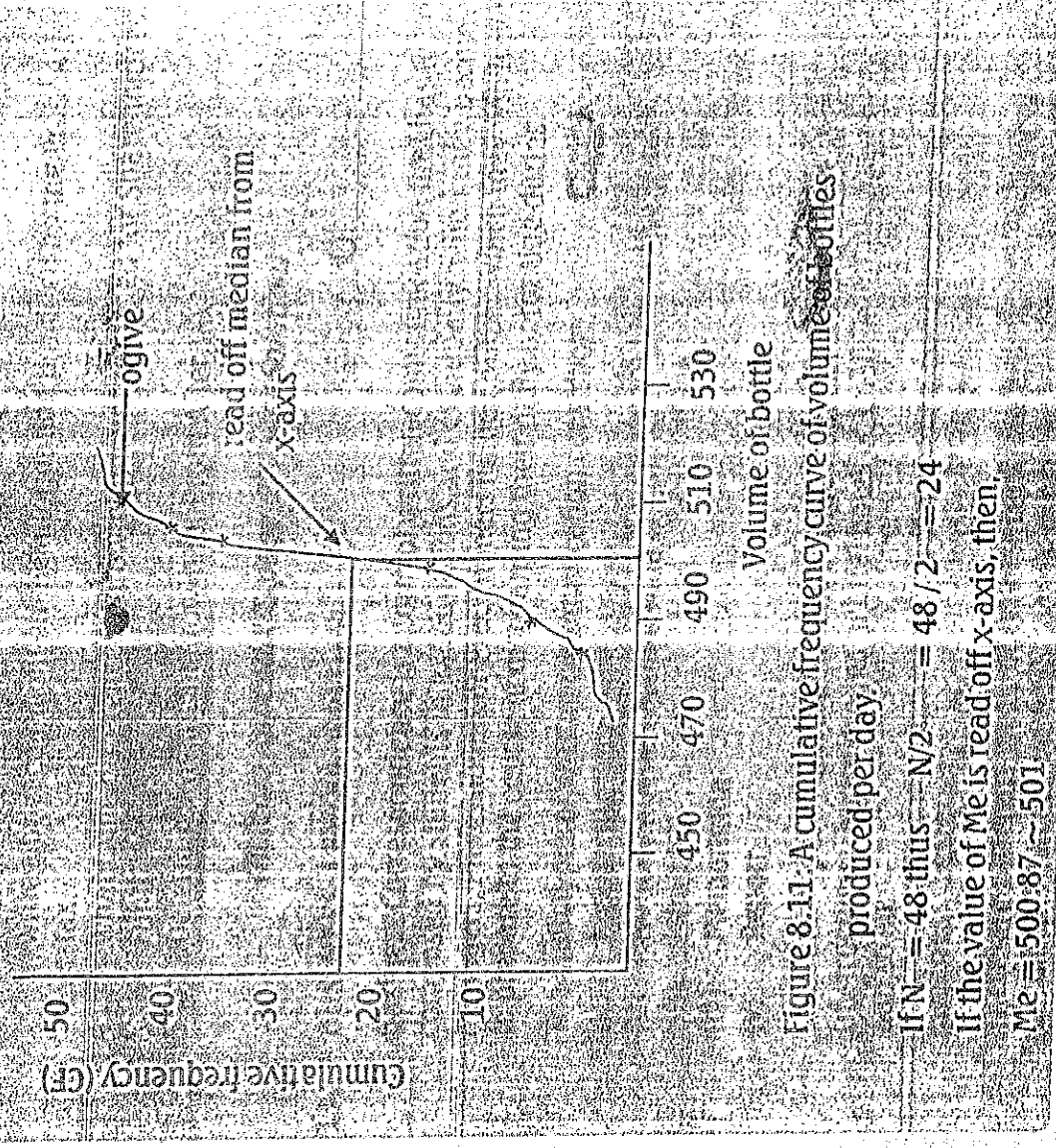
$$Me = 501.82$$

Graphical Method

When using the graphical method to calculate the median, the following procedure is followed:

- ◆ Determine the cumulative frequency.
- ◆ Construct a graph of c.f. against the class interval to form an ogive
- ◆ Determine half ($\frac{1}{2}$) of the number of observation or frequency that is $N/2$
- ◆ Locate the result of $N/2$ on the cumulative frequency axis.
- ◆ Trace the point on c.f axis to the curve (Ogive) and then trace it down to x-axis and read off the value from the x-axis. This value gives the median.

Using the illustration 8.12 a graphical presentation can be made as follows:



(c) Mode

The mode is an estimate of the value that occurs the most in a set of data. That is, the value that has the highest frequency. Mode can be used for both quantitative and qualitative data analysis. For instance, if a researcher conducts a simple survey on peoples' opinion on a certain product in the market, the researcher can determine the mode qualitatively by simply noting the respondents opinion, which is either 'for or against' the product.

Computing the Mode

- (a) For Ungrouped Data

We simply look out for the most occurring frequency value. For instance in a set of data;

60 71 80 45 60 56 60 55 45 48 60 71 55 82 60
 The mark, which occurs the most, is 60, occurring five (5)-times. Thus, the mode of the data above is 60.

NB: Where two figures have the same frequency, then these figures are referred to as Bi-modal values.

- (b) For Grouped Data;

The modal class is determined and the formula below is applied:

$$\text{Mode} = L_o + \underbrace{\frac{F - F_b}{(F - F_b) + (F - F_a)}}_C \text{ — equation 8.5}$$

Where L_o = lower class boundary of the modal class
 F = actual frequency of the modal class
 F_a = frequency of the class just after the modal class
 F_b = frequency of the class just before the modal class
 C = size of the modal class.

ILLUSTRATION 8.13

Use the group data obtained for the volume of bottles manufactured per day in a company to determine the mode.

Volume of bottles	Frequency (f)
485 - 490	1
490 - 495	3
495 - 500	12
500 - 505	22
505 - 510	8
510 - 515	2

Determine the modal class, which is 500 - 505 since it has the highest frequency of 22.

Thus, using equation 8.5

When $L_0 = 500$, $f = 22$, $F_0 = 12$, $F_0 = 8$, $c = 5$

$$\begin{aligned} \text{Mode} &= 500 + \frac{22 - 12}{(22 - 12) + (22 - 8)} \times 5 \\ &= 500 + \frac{10}{10 + 14} \times 5 \\ &= 500 + \frac{24}{24} \\ &= 500 + 2.083 \end{aligned}$$

$$\text{Mode} = 502.08$$

ILLUSTRATION 8.14

Given the following distribution, estimate the mode.

Class Interval	Frequency
36 - 40	3
41 - 45	15
46 - 50	13
51 - 55	11
56 - 60	6
61 - 65	2

Table 8.13: Table of Estimated Distribution

SUGGESTED SOLUTION 8.14:

The modal class is 41 - 45 with the highest frequency of 15. Applying equation 8.5 gives

Where $L_0 = 41$, $f = 15$, $F_0 = 3$, $f_0 = 13$, $c = 5$

$$\begin{aligned} \text{Mode} &= 41 + \frac{15 - 3}{(15 - 3) + (15 - 13)} \times 5 \\ &= 41 + \frac{12}{12 + 2} \times 5 \\ &= 41 + \frac{12 \times 5}{14} \\ &= 41 + 0.857 \times 5 \\ &= 41 + 4.285 \end{aligned}$$

$$\text{Mode} = 45.285, \sim 45.29$$

Conclusion:

This implies that the frequency that appears the most in the set of grouped data is 45.

ii. Graphical Presentation:

This is very easy to obtain. The following procedure is used;

- ◆ A histogram is constructed for a given frequency distribution.
- ◆ The modal class is determined on the histogram. That is, the tallest bar.
- ◆ A line is drawn from the pick adjacent corners of the rectangular bar of the modal pick to the left pick and to the right pick rectangular bars next to the modal class.
- ◆ A line is drawn to x-axis from the point of intersection of the lines drawn from 3.
- ◆ The value of the mode is read off from x-axis.

ILLUSTRATION 8.15

Using the frequency distribution in example 14

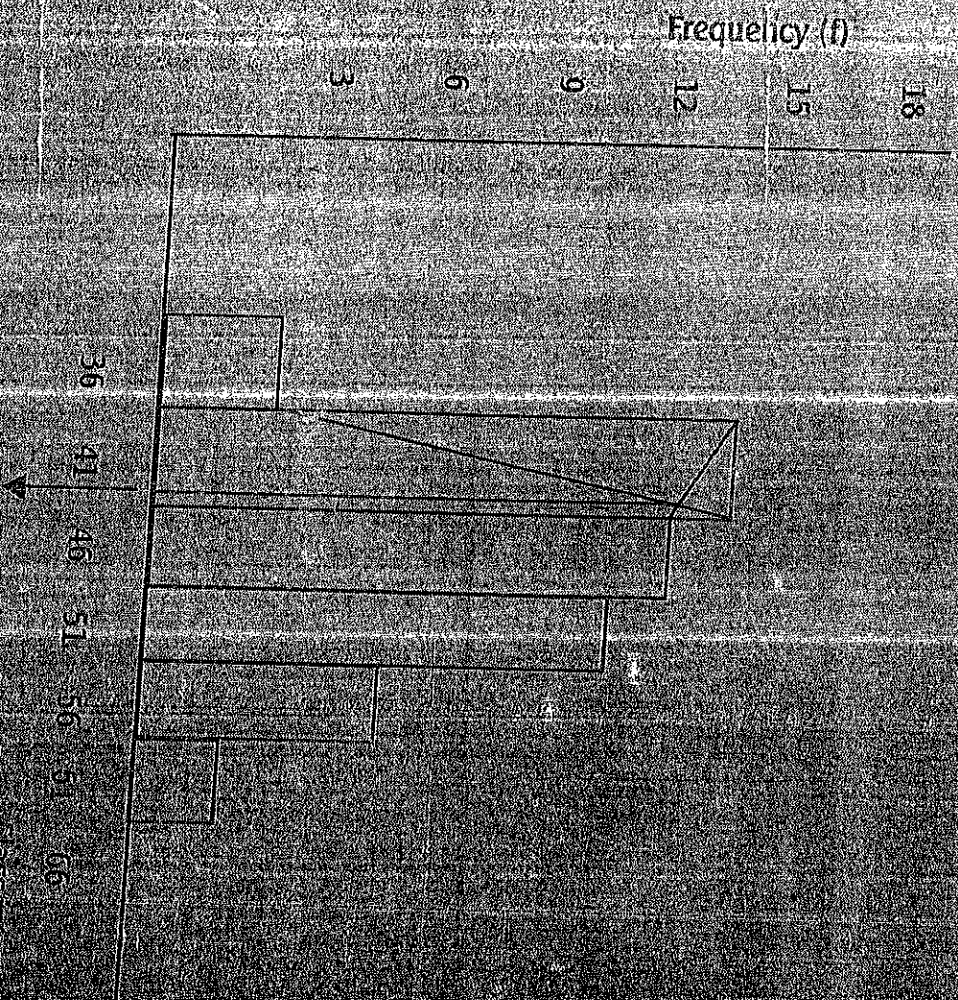


Fig. 8.12

A simple histogram

THE RESEARCH REPORT

12.0

LEARNING OBJECTIVES

Upon the completion of this chapter, readers should be able to:

- ◆ Identify the attributes of a good report.
- ◆ Plan and arrange any written correspondence in a logical order.
- ◆ Write a comprehensive interpretation and presentation of study's findings.
- ◆ Draw inferences and report such in their study report.
- ◆ Develop and write recommendations for further research on the findings.
- ◆ Define and identify differences between the terms, references and bibliography.
- ◆ Identify the differences between in-text and reference list.
- ◆ Identify the differences between the four types of referencing methods discussed.
- ◆ Write an acceptable format for referencing a text, and
- ◆ Define the use of appendices in a research report or paper.

12.1

INTRODUCTION

Writing a report, either in business, by a manager or any other senior person to whom the work is delegated or in education by a researcher for academic or official purpose is inevitable. In fact, in examinations, students may be required to write a report on a chosen subject or on a topic. All these call for systematic approach and techniques to be followed if a good report must be produced.

In business reports consisting of various forms of correspondence such as, letters, memos, minutes, notices, instructions, summaries and others, sufficient information must be provided or data collected in the best possible and economical way. Also, for effective communication to be achieved with these various forms of correspondence, there must be clarity of expression or plainness, conciseness or precision, consideration or simplicity, sincerity, accuracy and style. (These will be fully discussed in the next chapter). There is need to plan and arrange any written correspondence in a logical order and to ensure that there is unity and coherence of the materials used. Most importantly, the writer should adhere strictly to the rules of grammar.

12.2 REPORT WRITING

A report is a formal statement of the result of an investigation or of any subject in respect of which certain information is required, and is written by a person or body charged for this purpose. Report writing is a specialized form of communication and it is of different types and kinds. Apart from the type of report required from students in examinations, which is purely for a specific purpose, the other types are research or survey reports such as those found in academic, and commercial or business environments. For examination and business purposes, there are short, long and very long reports that range from simple and fairly complex to those technical in nature.

A good report must therefore be completely readable and formal in tone. The language used must conform to the expected standard and style or custom of the body concerned.

In business, the reports require careful planning in terms of selection, collection, organization and presentation of data. Reports however, may be classified into letter-form, memo form, schematic or mixed form. The schematic or paragraph form depending on its purpose, length and requirements.

Reports, as already stated, comprise informal – verbal and formal – non-verbal reports. The former is given at meetings, interviews or on the telephone as conversations while the latter is used also at meetings or for any other formal presentation as a written document. The various kinds comprise ordinary (routine) and special reports. Some examples of the formal reports are made by travelling salesmen, heads of departments, committees or at important occasions such as general meetings of companies. A special report is meant for special situations given in special circumstances. This type of report is made occasionally for specific purposes. For instance ‘an eye – witness account type of report’. Also there is investigation report that a senior person in an organization may be asked to write.

Styles in Research Reporting:

Seybo in Anao and Uche (2002) states that there are two widely used styles in research reporting which involve project writing be it academic or commercial (business) purpose. These are:

- ◆ The American Psychological Association (APA) style often called, Harvard style.
- ◆ The Kale Turabian style of research reporting often referred to as the Chicago style.

In their claims, it provides for clarity of purpose. Research report in academics involves dissertation, theses and journal articles or paper writing. It usually follows a selected system of format; it refers to the general pattern of organization and arrangement of the report. This must follow the rules or styles of the faculties, departments, research institutions and professional bodies. The researcher should find out, which format is acceptable by the body that he intends to submit his report to before writing the report.

12.2.2.1 Attributes of a Good Report

Report writing, whether it is the research type of report or any other type of report entails many attributes that make it readable and understandable. In other words, a good report must have some qualities or characteristics embedded in it for comprehension sake. Thus, it must be presented in terms of:

A. Layout: This should:

- ◆ Carry a title and date,
- ◆ Have a table of contents so that the information that the decision maker wishes to examine closely can be located without difficulty,
- ◆ Contain a clear statement of the objective of the research, and an explanation of the methods,
- ◆ Have a statement of findings and recommendations in non-technical language appearing at the beginning. If findings are extensive, they should be summarized.
- ◆ Have the main body presented in logical sequence and broken into readily absorbed and identifiable sections and
- ◆ Have the technical data, for instance, sample information or that of questionnaires and statistical tables put in an appendix.

B. General Rules;

- ◆ The report should be written in a concise and objective style, avoiding vague expressions such as 'it seems', 'there may be' and the likes.
- ◆ The report writer should seriously consider the inclusion of charts and illustrations in the main body of the report if they present information more clearly or forcibly.
- ◆ The report writer should state any constraint or calculated confidence limits.

12.2.2 Presenting the Findings of the Study

The data collected have to be presented when the report is written. The first step in examining a body of data is to present them in a form, in which the main features can be readily appreciated – a graphical presentation is often very useful for this purpose. The choice of a diagram will depend on the type of data and on the features, which the diagrams are required to show. The different types of diagrams that can be used to present data include;

pictograms, pie charts, bar charts, line charts and so on. Hassan (1995: 268) states that a good report presents the findings of a study in a clear and detailed manner such as to permit another investigator to replicate it without difficulty.

As a scientific document, the research report must not contain subjective statements. The researcher must not set out to defend a portion. He should, however, report the findings as an unbiased assessor.

Typically, a research report should contain only objective, factual description of past research and the outcome of present endeavour. The report should be written in simple and straightforward language. The language should be clear, unambiguous and scholarly.

Errors of spellings, grammar and punctuation should be avoided. Researcher should avoid the use of personal pronouns such as 'I', 'my', 'we' and so on. Rather, impersonal pronouns and passive voice should be used.

The outline in the APA publication manual provides the format for presenting a research report, which consists of the title page, abstract, introduction, methods, result, discussion, references and appendix (Where applicable).

In research report writing, chapter five, which comes last in a research project, presents the findings of the research or the study, the conclusions drawn from the data analyses and recommendations made. If this section on findings is not properly presented, the entire work will be damaged. There should however, be a connection between chapters 1, 4 and 5.

12.2.3 Interpreting the Findings

Research is very useful for planning and decision-making. As a result, various reasons can be adduced for undertaking a research. The first reason, which is for policy analysis, involves validating hypotheses that require inferential statistics. The second reason is that research is also undertaken in order to obtain estimates about the magnitude of relationship requiring descriptive statistics. The last reason may be for the purpose of forecasting.

Soyombo in Anao and Uche (2002) refers to the interpretation of research findings as a key component and one of the most important aspects of the research process. A research process becomes useless to policy making or planning if its outcome cannot be properly analyzed and interpreted. The results of the findings must be presented in a meaningful way to the reader of the report by using appropriate statistical methods. The policy implications of the research findings must also be highlighted and well articulated. Empirical research is often conducted with a purpose of influencing policy either in the short or long run.

For the purpose of analysis and interpretation of research findings, the users of such findings must be made to understand the important indicators and statistics reported in research findings and how to use them in interpreting the research findings

The important keys to look for in interpreting research statistics are:

- ◆ The simple correlation coefficient, (r) to determine whether or not there is a relationship between two or more variables.
- ◆ The coefficient of determination (r^2) to determine the extent of the relationship among variables using regression analysis.
- ◆ Regression coefficients - These are the measures of the respective impact of changes in the explanatory changes on the dependent variable.
- ◆ The F-statistic - This is used to test whether the estimated coefficient of determination is statistically significant that determines the extent of its reliability.

The key also states that the following principles are quite important in interpreting research findings. The researcher:

- ◆ Must have at the back of his mind the target audience
- ◆ Should understand the concepts and terms used in the analysis.
- ◆ Should appreciate the relationships between the variables whose determinants are to be explained.
- ◆ Should understand the methodology adopted in carrying out the research and its implications.
- ◆ Should know the appropriateness of the model to the real life problem being studied as contrary to the above stated facts.
- ◆ Must be certain of the theoretical appropriateness of the findings and the admissibility of the data.
- ◆ Should know that the research findings must be interpreted in the context of the environment of the researcher.
- ◆ Should know the interpretations and must be internally consistent.
- ◆ Must also make allowance for alternative interpretations of research findings.
- ◆ Should contextualise the results in relation to other previous studies in the area of investigation if they exist.
- ◆ Should clearly point out the policy implications that flow from the research efforts he embarked upon.
- ◆ Should carry out the interpretation of the research within the context of the objective of the study.

12.2.4 Methods of Presenting the Findings of the Study

The most popular methods of presenting the outcome of a research comprise visual representations by means of line graphs, pie charts, tables and so on. The others include correlation coefficient, the standard error and the

coefficient of determination. Whereas in descriptive analysis, the ratio, percentages and trend analyses among others, are also used to present research outcome.

The use of tabulation for instance, makes the quantitative data that have been coded to be simply presented for clarity of purpose and better understanding possible. This involves the use of tallies, machine processing and so on. Also, statistics such as, parametric and non-parametric statistics, which include percentages, mean, standard deviation and correlation coefficient are based on random samples. These involve the distribution of the variables in the population, which are also employed during the course of research study and presented in the findings.

12.2.5 Drawing Inferences and Conclusion from the result of the Study

The result of the study can be generalized by making some inferences from the just concluded research. With inferential statistics, this is quite possible, rejection of the hypotheses. Based on this, conclusion can be drawn and recommendations made. For instance, inferential statistics helps to infer meaning and important relationships, which are hidden within the data.

Asika, (1991:128) states that the main functions of inferential statistics are as follows:

- To estimate a population parameter from a representative sample drawn from the population.
- To predict the population characteristics from the characteristics of a randomly selected sample of the population.
- To test relevant hypotheses for the purpose of drawing valid conclusions for research studies.
- Conclusion on the other hand, are deductions from the findings and should be drawn from chapter 2 (Review of literature) and chapter 4 (analysis of findings).

Every conclusion must be relevant to the findings of the study and should not be extraneous to the study. Some students often try to draw conclusions on their study from their personal opinions which cannot be defended with the findings from any part of the study. It is advisable for students to eschew this practice.

12.2.6 Developing Recommendations and Signpost for Future Research from the Findings

In chapter 5 of a research project, the summary of the findings and the conclusions drawn from this as well as recommendations made are presented. Every research report must have objectives and focus. It is in the interest of the researcher to provide some recommendations based on the

conclusions drawn from the findings. The researcher may also make recommendations for further studies or research to guide future researchers, who may wish to embark on similar projects in order to expand the frontiers of knowledge.

In this respect, the researcher recommends to the 'would be researchers' some aspects of his study or the results of his study, which can lead to further research by any interested researcher. Thus, researchable topics are made as suggestions for possible areas in which empirical studies can be carried out.

Some researchers have found this part very useful in selecting researchable topics. The advantage of using this recommendation for any study is that the researcher has a basis already, which will guide him in his own study. He can then draw from the literature of the parent study. He may use the study's method of statistical analysis, adopt its methodology or if need be, replicate the study. (Azika, 1991:108). However, for the sake of creativity and originality, it is usually advisable for the researcher to develop his own methodology. For instance, he may decide to conduct a research study on similar topic to that borrowed from existing project report, but use a different group of population, methodology and statistical analysis. This, however, provides a new set of idea about the study thereby adding to the advancement of knowledge in that area of study.

12.2.7 Executive Summary

Again, in the last chapter of the research report, there is the summary section, which has the summary of the findings and of the entire study. Here, the researcher tries to state the essential and revealing aspects of every chapter. Brevity of the explanation is normally required in this case also.

For technical research project of commercial (business) type, the format includes an executive summary. This is in form of an Abstract. It is referred to as the epitome of salient facts and important points that must not be left out. The summary provides the reader with a quick grasp of what the research report is all about and saves him some valuable time, which could have been spent in reading the entire report.

12.3 REFERENCES AND BIBLIOGRAPHY

What is Reference?

This is a standardized method of acknowledging sources of information and ideas that are used in an assignment, term paper, project research, thesis, dissertation, article and other academic write-ups in such a way that it uniquely identifies the sources.

Direct quotations, facts and figures, ideas and theories from both published and unpublished works must be referenced or acknowledged.

A reference list at the end of the assignment contains full details of all the in-text citations that are cited in the body of the write-up.

What is Bibliography?

This is a standardized method of acknowledging relevant sources of information and ideas that are used in the body of the work. Such sets of information or ideas may have been obtained through casual observation or experience, which are relevant to the topic at hand.

12.4 REASONS FOR REFERENCING

There are three basic reasons why it is necessary to reference write-ups; these are:

1. To avoid plagiarism.
2. To enable the readers verify quotations.
3. To enable the readers do a follow-up and read more fully the cited authors' views.

12.5 TYPES OF REFERENCING

There are many acceptable types or forms of referencing. This text deals with the APA (American Psychological Association), Harvard, Chicago and Library of Congress methods. At this point, it is important to note that the APA and Harvard methods of referencing are similar in many respects. The main difference lies in the use of punctuation marks, which vary from one research body or organization to another, with the standard format or procedure usually being the same.

12.6 STEPS INVOLVED IN REFERENCING AS IDENTIFIED IN APA AND HARVARD FORMATS.

The following elements are contained in the APA and Harvard format.

A. For books:

- (i) Author or editor
- (ii) Title of book
- (iii) Edition
- (iv) Volume number (if any)
- (v) Place of publication
- (vi) Publisher

B. For journal articles:

- (i) Author of article
- (ii) Year of publication
- (iii) Title of article
- (iv) Title of journal
- (v) Volume number
- (vi) Issue number
- (vii) Page numbers on which the article appears

C. For electronic information:

- (i) Author of publication
- (ii) Year of publication
- (iii) Title of article or book
- (iv) Title of journal (not necessary for books)
- (v) Type of medium (CD-ROM, motion picture and so on)
- (vi) Pages or length (6-14, 71 p, 25 Paragraphs)
- (vii.) Retrieved statement (www address, name of electronic Database, E-mail address and so on)
- (viii.) Access date.

The reference list is provided at the end of the document while the citations are inserted at the appropriate place or paragraph of the cited reference within the text of the document.

NB: It is necessary to note that in most cases, all of these details may not be applicable or even available.

12.7 HOW TO CREATE A REFERENCE LIST

A reference list contains all the references cited in the body of the assignment, unlike the bibliography, which cites works that were consulted or recommended for further reading.

The reference list is arranged alphabetically using the authors' surnames. Where there is no author cited (Although in rare cases), it is cited by its title and ordered in the reference list or bibliography in sequence by the first significant word of the title.

The APA and Harvard styles require that the second and subsequent lines of the reference be indented.

ILLUSTRATION 12.1

(A) For Books:

Cozby, P.C. (2000). *Methods in Behavioural Research*. California: Mayfield Publishing Company.

When the reference is written in print, the title of the book can either be typed in **bold** or *italicized* in an examination condition, where pen is used it may be underlined. For books with more than one author, the names of all the authors are written. For instance,

Anao, A.R. and Uche, C.U. (2002). *Research Design and Implementation in Accounting and Finance*. Nigeria: University Of Benin Press and Institute Of Chartered Accountants Of Nigeria.

(B) For articles or chapters in a book or an encyclopedia:
Bergmann, P.G. (1993). *Relativity*. The New Encyclopedia Britannica. (Volume 26, PP501-508). Chicago: Encyclopedia Britannica.

Ogunloye, A.O.O. (1999). *Improving The Quality Of Educational Research* in

Obel, E. O University of Lagos- Lagos series in education- crucial issues in Nigeria Education. (Volume 4, PP 132-147). Lagos: Amosound press

(C) For articles in journals, magazines or newspaper: examples are,

Egusa, Imade (2004). *Marketing Financial Services*. The Nigerian Accountant- Institute Of Chartered Accountants of Nigeria. (Volume 37, no 1) pages 34-35.

Otiaku Kalu (2004, December 11). *ASet Up A Fast Track System For Manufacturers*. The Comet Newspaper. (Volume 6, PP 27 No.1949).

NB: The volume can also be written as (1997) or (1949/6). To reference articles published by the same author in the same year, the same format above is adopted except that the date of publication is tagged as 'a' or 'b' for each article or item.

12.8

HOW TO REFERENCE ELECTRONIC TEXTS

ILLUSTRATION 12.2

(A) For full text journal article from CD-ROM(BPO). An instance is:
La Rosa, S.M. (1992). *Marketing Slays the Downsizing Dragon*. Information Today, 9(3), Pg 58-59. Retrieved January 15, 1999, from [LWI/Business Periodical On disc](#).

(B) For full text journal article from an internet-only journal.

Fredrickson, B. L. (2000, March 7). Cultivating Positive Emotions to Optimize Health and Well-being. *Prevention and Treatment*. Article 0001a. Retrieved November 20, 2000, from <http://journals.apa.org/prevention/volume3/pre0030001a.htm>

(C) For CD-ROM (stand-alone Multimedia Disc)

SPSS 11.0: Data Editor, Entering Non-Numeric Data (2004). Retrieved December 19, 2000, from SPSS 11.0 software

(D) For World Wide Web pages (www.pg.jhu.edu/~withno/authoring-america-for-the-21st-century-developing-a-strategic-plan-for-educating-leadership-for-columbia-university-1993-2000-final-workshop-draft) (1994). Retrieved May 16, 1995, from <http://www.ilcolumbia.edu/CONF7edplan.htm>

NB: The same format is applicable for www, single author, joint author or institutional publication with or without publication date. The retrieval date is of utmost importance here.

(E) For ERIC Document (microfiche) (Education Research Information Center)

The reference text here requires a document number while all other sequences remain the same. For instance:

Davis, R. K. and Lombardi, J. P. (1996). The Quality of Life of Rural High School Special Education Graduate: A Follow-up Study. In *Rural Goals 2000: Building Programs That Works*. [microfiche]. (ERIC Document No. ED394765)

(F) For Discussion List

Simons, D. J. (2000, July 14). New Resources For Visual Cognition. [Msa31]. Message posted to <http://groups.yahoo.com.au/group/visualcognition/> message/31

(G) For video recording and films. This needs the name(s) of the principal contributor(s) with their function in round bracket. The medium is indicated in square bracket after the title, and the location and name of the distributor. For instance:

Young, T. (Director), and McClory, K. (Producer). (1985). *Tunderball*. [Motion picture]. USA: Warner Home Video

12.9 GOVERNMENT AND PARLIAMENTARY PUBLICATIONS

These appear in three major folds, which are Legislation, Acts of Parliament and Cases.

Legislation:

This is included in the reference list of bills or laws that are used, where they are important to the work. Thus, it is best to set the list apart from the main body of the reference list and present it alphabetically under a sub-heading titled, 'Legislation'. The same format is adopted for referencing. If legislation is obtained from an electronic database, the retrieved statement is added as it is for electronic journal articles retrieved from an electronic database.

Acts of Parliament

This has short formal title that can be used for citation purposes. The first time the Act is cited, gives the short formal title in italics. In subsequent references, the title is written in Roman type and the date is omitted. For instance: Title, Year of publication. Section, Subsection and Item (if any). For instance

Road Traffic Act 1972, S. 5 (2) (a) that is
Title, Year of publication and Schedule
OR

Transport Act 1981, Sch. 8

Cases

The name of the case is italicized and the year is placed in bracket. Round bracket is used for general year while square bracket is used for very important year.

For citations, the family name of the first-mentioned party on each side is cited.

For corporation, their names are usually stated in full (although abbreviations may be used as stipulated by the institute). For instance, the following sequence is adopted.

Parties, Volume/Report, Year, Call page @ particular page per Judge & Court
An instance of this is:

Hall vs Tea, 27NWL.R. (1999).36 @ 39 PER Sowemimo, J.C.A

This implies that the case between Hall vs Tea, in volume 27 Nigerian Weekly Law Report, of 1999 from page 36 at exactly 39th page per Judge Sowemimo, in the Court of Appeal.

For a case with very important year;
OR

Jamieson vs Cooper, [1942] 67ELR, 316

Tifau vs Department of Labour, [1984] 2NLR, 396
NB: Where ELR is European Law Report.

12.10 CITING REFERENCES WITHIN THE TEXT OF WRITE-UPS

When citing references within the text of a write-up, the general format is to use only the surname of the author and the year of publication in round bracket. The relevant page numbers must be included, when quoting directly from the sources. For instance; Osuala (1993:23) states that If there are no page numbers, for instance in electronic sources, the paragraph number is used. If neither is given, then the year alone is used.

12.10.1 Citing Work With No Author Or Multiple Authors

For a piece of work with no author or legal materials, the first few words of the reference list entry (usually the title) and the year are quoted. Double quotation mark around the title of the article, chapter, journal, book, brochure or report is used. For instance; On free care.....

("Study found" 1982) or "The Book College Bound Seniors" (1979). For a piece of work with multiple authors, where the names appear as part of the text, then they are joined by the word 'and' but where the names occur within parentheses, then an ampersand (&) is used to join the names. The year of publication is also included. Where the authors are more than two (2), the surname of the first author is written followed by 'et al' and the year of publication.

12.10.2 Citing From Web site and Secondary Sources

To cite an in-text from web site, the address is given in round bracket for instance: MetaCrawler(<http://www.metacrawler.com>) is a meta-search tool Include thereafter, the page, but where there is no page or part of that site, then it is not necessary for its entry in the reference list.

For secondary sources within a text, the original source is named as well as a citation for the secondary source. For instance; Akande and Bayelo study (as cited in Obe, 1999)..... For the reference list, the secondary source alone is referenced.

12.10.3 Personal Communication and E-mails

This includes; letters, memoranda, conversations and electronic mails. These are categorized as bibliography since they may not be entered into the reference list but an in-text citation is required. The initials and surname of the communicator as well as the date are included. For instance;

Peters specifies that the court wherein the judgment of the case was made be stated. (Personal communication, December 26, 2004).

OR

The company's policy on refunds was contrary to that of state law (L. Noel, personal communication, May 2, 2000).

12.1.1 CHICAGO STYLE OF REFERENCING

This method makes use of a simple procedure called the 'footnote and endnote'. The former comes at the bottom of the page while the later comes at the back page and it is referred to as reference list and (or) bibliography.

It uses 'ibid' (in the place of) as a method of referring to the work, which has been previously referenced in another book or in-text. It also encourages a method of simplified referencing, which allows the researcher or writer to use an abbreviated form in the bibliography. It uses both a reference and bibliography page and the lists are numbered as 1, 2, 3, ...n.

Steps in Referencing

The following sequence is adopted for this type of referencing.

A. For Books:

- (i) Name of the author
- (ii) Title of the book
- (iii) State / country of publication
- (iv) Publisher
- (v) Year of publication

Example of this include:

1. Dick Francis, *Come to Grief* (New York: Putnam's Sons, 1994).
2. Liam P. Unwin and Joseph Galloway, *Peace in Ireland*. (Boston: Stronghope Press, 1990).

B. For Journals:

- (i) Name of author
- (ii) Title of article
- (iii) Title of journal
- (iv) Volume / issue number (if any)
- (v) Year of publication
- (vi) Page(s) of article.

ILLUSTRATION 12.3

3. Rush Limbaugh, 'The Best of the Reagan Years', *National Review*, 29(1990), 22-25.

C. For Newspaper article:

- (i) Name of the author
- (ii) Title of the article
- (iii) Title of the newspaper
- (iv) Date
- (v) Section / page

Example:

4. Williams Robbins, "Big Wheels: The Rotary Club At 75", New York Times, Sunday, February 17, 1980, Sec.3

D. For Interviews:

- (i) Name of interviewee
- (ii) Name of interviewer
- (iii) Place of interview
- (iv) Date of interview

Example:

5. George Bush, Interviewed by Dan Rather, CBS Evening News Columbia Broadcasting System, February 8, 1989.

E. For Document:

- (i) Name of author
- (ii) Title of document
- (iii) Place found / resource center
- (iv) Date of publication
- (v) Identifier number
- (vi) Page(s) found

Example:

6. Judith Glazer, "The Master's Degree Tradition, Diversity and Innovation", In ERIC Education Resource Information Center, (1986)
Identifier number ED279260 (22-23)

F. For Internet:

- (i) Author(s) name(s)
- (ii) Title of book/article
- (iii) Place of publication / address
- (iv) Date cited / retrieved date
- (v) Web site

Example:

7. LS Bixley, Sentient Microfilm: A Tempest In A Test Tube
[Available online] Princeton University, cited January 1996.
Available from <http://www.princeton.edu/pub/harvard/psyc/92.3.26/consciousness/11.html>

12.12 DIFFERENCES BETWEEN REFERENCES AND BIBLIOGRAPHY

References consist of all documents that are cited in the text of the manuscripts.

These include journals articles, books, and chapters in books, technical reports, computer programmes and unpublished work that are mentioned in manuscripts.

Thus, reference is therefore a standardized method of acknowledging sources of information and ideas that have been used in an assignment in a way that it uniquely identifies their sources. Direct quotations, facts and figures as well as ideas and theories, from both published and unpublished works must be referenced.

Bibliography is often confused with or interchangeably used as references but they differ from one another. For instance, a bibliography contains everything that would appear in the reference section including other publications that are useful but are not cited in the manuscript. However, it is common to find references rather than bibliographies being used in research reports.

References are listed in alphabetical order by the surname of the first-named authors. When no author is listed, the entry is stated with the word of the title of the article or the sponsoring organization. Each reference starts at the left margin of the page, with subsequent line double-spaced and indented. No extra spaces separate the entries.

Bibliographic details are sectionalised and arranged in sequence such as books, journals, articles newspapers, magazines, unpublished work, Eric document, Personal Communication E-mails and so on. A bibliography cites work for background and further reading whereas references list contains all the references cited in the assignment.

While the bibliography lists all works (Books, journals magazines, newspaper and so on) which the author used, whether or not such works are actually referred to in the main text, references refer to all work that are actually used by the author in the course of his writing. These must be stated in such a way that it would not be difficult for readers or users to trace or locate the sources from which a researcher has cited.

Bibliography must appear at the end of the project and in fact a bibliography is always the last portion of a book or project but it is sectionalised or grouped under different headings such as books, journals, articles unpublished work and so on.

Thus, bibliography is a type of referencing that is highly patterned as it lists all relevant materials consulted in their significant order and arranged together

Asika, N. (1990), however, says that although a bibliography looks like end-of-chapter references, it differs from the latter in some ways. These, he stated as follows:

- ◆ A bibliography appears only once in the entire work and at the end of the project. (End-of-chapter references appear at the end of every chapter)
- ◆ In bibliography, the author's last name is written first before his other names and / or initials. (In end-of-chapter references, the author's first names precedes his last name such as; in Kate Turabian style only)
- ◆ In a bibliography, all works consulted by the author are arranged as follows: Journals, books, magazines, newspapers and finally other publications. In end-of-chapter references, this arrangement is not necessary.
- ◆ In a bibliography, all the works for each of the sections, for instance, journals, books magazines and so on are arranged alphabetically and if necessary numbered serially. While in end-of-chapter reference, the serial reference notations determine the locations of the references.

According to Obilade (1987), referencing a research work, whatever style one uses, must be consistent. This is because while some requires footnotes, others discourage their use. Some prefer a bibliography, others require references.

Some want reference note or reference list at the end of each chapter. The reference list (called 'References') at the end of the study lists all the sources that are used in the text alphabetically. The reference should be as complete as possible. It should include information on author, title and publication. Generally, the footnote is not needed if enough information has been provided in the text.

Obilade (1987), however highlighted some important points that are needed to note whenever a particular style of referencing is chosen. These are;

- (a) All works cited in the text must be included in the bibliography or reference.
- (b) The reference or bibliography must be alphabetically arranged with surname first.
- (c) The bibliography or references must not be numbered
- (d) In the case of multiple authors, the name of all the authors must be included.
- (e) In the case of references such as, dictionary and/or encyclopaedia, the work (and not the editor) is arranged in alphabetical order.
- (f) In the case of the unsigned article and anonymous works, the first content of the words are arranged in alphabetical order.

Similarly, Aromolaran (1998) is of the opinion that the portion on references and bibliography is very important in any scholarly and publishable article. It is very rewarding for authors, whose materials have been used to be duly acknowledged. He further added that references are mentioned for instance as [Aromolaran (1996),.....]. While bibliography is necessary for instance texts, articles and so on, are found to be useful in the course, references are not made of the authors in the articles.

Bibliography is also necessary if the name of an author or his surname with his initials (For instance Boye T. Adewale (1997) or B.T. Adewale (1997) is of the opinion that.....) are stated. Both references and bibliography must be well written in a standard format.

12.13 LIBRARY OF CONGRESS APPROACH OF REFERENCING

Over the years, the Library of Congress approach to forming references have been by manual organization of resources, which ranges from texts, encyclopedias, newspaper reports, maps, films and various antique objects.

In time past, this approach makes use of the 'Machine Readable Catalogue' (MARC). This approach is a widely used standard for the presentation and exchange of resources. These resources include: authorities' reports, bibliographic information and classification, community information and holdings of data processing units. All these are now being presented in machine-readable form and the Library of Congress Association in five (5)-coordinated formats presents them. Available online, <http://www.loc.gov/marc/specifications/>

The MARC format provides a quick reference guide to the content defined in each coordinated format. It provides a concise description of each field, character position of the fixed-length data element fields and of the defined indicators in the variable data field.

Codes and coded values are used, when their names may or may not be sufficiently described in the various sub fields.

The Library of Congress approach has employed a 'test-in-front and image-behind presentation of references, which is made available online. This is appropriate when the accuracy of the converted text is reasonably high and when the text is formatted sufficiently to permit a comprehensible display.

In the test-in-front approach, once the researcher has identified the item, which he or she wishes to view by searching the bibliographic records for a collection, the item is presented as a Hypertext Makeup Language (HTML) and or Standard General Makeup Language (SGML) version of the text.

The library encodes its document usually by using the Standard Generalized Markup Language- (SGML; ISO 8879). ISO being the International Organization for Standardization. - <http://www.oasis.open.org/cover>

a. **Components of MARC:**

It comprises three (3) elements: record structure, content designation and data content.

Record Structure:

This is an implementation of the American National Standard Interchange, which is coded as (ANSI / NISO Z39.2)

Content Designation:

This gives the tag, code and conventions that are established to identify and further characterize the data element within a record. More so, to support the manipulation of data that is defined by MARC.

Content Data:

This is usually defined by standards outside the format, such as International Standard Bibliographic Description (ISBD), Anglo-America Cataloguing Rule 2 edition (AACR 2), Library of Congress Subject Headings (LCSH), Holding Statement Summary Level (ISO10324), Library of Congress Classification (LCC) and American National Standard for Serial Holdings Statements (ANSI / NISO Z39.44)

b. **Features of MARC:**

MARC records consist of three (3) main features: the leader feature, the directory feature and the variable field feature. [Retrieved from] <http://www.loc.gov/marc/specifications/>

Leader Feature

This contains data element that provides information for the processing of the record. These elements contain number or coded values, which are identified by a relative character position. The leader is fixed in length at 24-character position.

Directory

This contains a series of entries that have tags, length and starting location of each variable field within a record. Each entry is 12-character position in length.

Variable Field

MARC data are organized into variable fields, each identified by a three (3)- character numeric tag that is stored in the Directory entry for the field. Each field ends with a field terminator character, [ASCII ID hex]

It has two (2) types of variable fields: variable control and variable data fields

Variable Control Field

It is tagged (00X fields); it has a structural difference from the variable data field and it may contain a single data element or a series of fixed-length data element, identified by relative character position. It does not have indicator position or sub-field codes.

Variable Data Fields

This is tagged [01X-8XX fields]. Within this field, two (2) kinds of content designations are used, these are;

Indication position

This shows where the data is located. For instance, the first two-character positions at the beginning of each variable data field have values, which interpret or supplement the data found in the field.

Sub-field codes

This consists of a delimiter such as [hex f] and a data element identifier, which are lower case alphabetic or numeric characters. They are defined independently for each field, say for where exceptions exist. More so, they are defined for the purpose of identification. The order of sub-fields is specified by standards for the data content, such as cataloguing.

<http://www.loc.gov/help/desk.html>.

<http://www.loc.gov/marc/concise/concise>.

<http://www.loc.gov/marc/bibliographic/ecb/home.html> and <http://www.loc.gov/marc>

12.14 THE BIBLIOGRAPHIC FORMAT FOR RECORDING DATA IN MARC

This is designed to be a carrier for bibliography information about printed and manuscript, textual materials, computer files, maps, music, continuing resources, visual materials and mixed materials.

Bibliographic data include;

- (i) Title of publication
- (ii) Name(s) of author
- (iii) Subjects
- (iv) Notes
- (v) Date of publication

- (vi) Information about the physical description of the item.
- (vii) Retrieval position or location

Data element of the following types is contained in bibliographic format:

- ◆ Books: textual materials that are monographic in nature.
- ◆ Continuing Resources (serials): textual items with a recurring pattern of publication, for instance newspaper, periodicals, yearbooks and so on.
- ◆ Computer Files: computer soft wares, numeric data, computer-oriented multimedia, online system or services (materials may be monographic or serial in nature)
- ◆ Maps: all types of cartographic materials, such as sheets maps and globes in printed manuscript, electronic and microform.
- ◆ Music: printed and manuscript music
- ◆ Sound Recording: non-musical sound recording and musical sound recording
- ◆ Visual Materials: projected-media, 2-dimentional graphics, 3-dimentional artifacts or naturally occurring objects and kits.
- ◆ Mixed Materials: primarily archival and manuscripts collections of a mixture of forms of materials. (Materials may be monographic or serial in nature).

a. Example of Library of Congress Bibliographic Format:

A general approach is used such that contains title, description and keyword resource.

For instance,

Samuel Demas, Peter McDonald and Gregory Lawrence, 'The Internet and Collection Development: Mainstreaming Selection of Internet Resources', Library resources and Technical Services 39:275-290 (July 1995).

NB: It is important to note that a specific outlook presentation is not provided; however a prescription of the APA or Harvard format is used.

12.15 LIBRARY OF CONGRESS APPROACHES TO MANUAL REFERENCING

This informs its users: students, researchers and others on how best to locate materials (Sources) from the Library. Fagbohunge (1993) listed three (3) sources from which information could be sourced in the library based on the understanding of their collections. These are:

- ◆ Card catalogue
- ◆ Indexes and Bibliographies
- ◆ Reference books

He is also of the opinion that the library staff could be assisted to locate books, journals, periodicals and pamphlets that are needed.

Card catalogue is an index, which lists all the publications in the library collection by author, by subjects, by title and by call numbers. It indicates its position on the shelf. The catalogue is alphabetically arranged just like the dictionary. Where a particular author has many books, all his or her books will be arranged alphabetically together under his surname.

The card catalogue does not cover all the materials available in the library. For instance periodicals and some important magazines for literature review may not be indexed in the card catalogue, yet the researcher needs these materials. For such materials that are not available in the catalogue, the researcher should be on his search light on periodical indexes and subject bibliographies. The library staff will direct as to their locations.

The reference books such as encyclopedias, directories, year books, general and special dictionaries readily come from the mind here.

As with the general character of these reference books, reference materials cannot be indexed in any specific way in the catalogue. Such reference books are placed in special locations within the library.

In some libraries, a special section called 'reference section' is created for them. If the library that is being used by the researcher has a chart indicating its sectional arrangement, the job becomes easier. But where this is not available, the library staff can direct the user.

Undoubtedly, a researcher would need the library for many reasons. These include:

- ◆ To gain the benefits of preliminary orientation.
- ◆ To seek published materials.
- ◆ To gain the background knowledge of the problem that is being investigated.
- ◆ To seek valuable and relevant information to the problem of investigation.
- ◆ To avoid unnecessary dissipation of efforts, energy and waste of time.
- ◆ To reduce costs.

12.16 APPENDIX

This forms a part of a research study report, which contains detailed information that are too long or would seem inappropriate for the main body of the report. In a research report, this portion is optional in the sense that the detailed materials or document may be provided as evidence where it is absolutely important or needed (For instance, the format of a prepared

questionnaire or relevant tables of figures or proof of a theorem used in the text) or there may be no need for such document in other forms of reports and as such the appendix will not be necessary. Thus, the appendix should be used in a research report where it is necessary.

- a. **Characteristic Features of an Appendix**
 - ◆ The following are some of the features of an appendix;
 - ◆ Each appendix should begin on a new page.
 - ◆ It should carry a label that is usually written in upper case alphabetical order tagged 'Appendix A' the next being B, C and so on.
 - ◆ It should have a centered title of the information on the appendix.
 - ◆ The information on the appendix is presented in its full form.
- b. **Examples of Items on the Appendix**
 - ◆ The appendix should include the following kinds of materials in its section.
 - ◆ Computer programs.
 - ◆ Unpublished research instrument that is originally constructed or copied from another source.
 - ◆ Lengthy treatment of cases that may not be available anywhere else.
 - ◆ Tables of all kinds, pictures, numbers and graphical presentation.
 - ◆ The cover letter that is used to convey or introduce the research instruments for instance questionnaire to the sampled respondents or organization surveyed as case study.
 - ◆ The research instrument used for collecting the data.
 - ◆ Proofs of theorem used in the text.

12.17 SUMMARY AND CONCLUSIONS

The aspect of research that deals with presentation of report is discussed within chapter. It presents various styles of writing research report as well as the attributes of a good report. The methods of interpreting, presenting, drawing up inferences and conclusions of the finding in a research study are also discussed.

The various methods of referencing the text used in a research study are presented with illustrations. These include; the APA – method, Harvard method, Chicago method and the Library of Congress method. Steps involved in referencing different types of text and materials used are discussed with illustrative examples. The differences between reference and bibliography are also discussed.

The concept of appendix is viewed by discussing its characteristics features and enumerating various items that can be listed on the appendix page of a project or research report.

(Refer to Comprehensive Questions and Solutions in Appendix II, page 295).